

ISPS Project 2019

Statistical Strategies for Global Warming Challenges



Pushkar Kale
Mrunmayi Naik
Chinmay Dabhade

Dept. of Statistics, Savitribai Phule Pune University, Pune.

ABSTRACT

The factors associated with global warming, ranging from increase in CO₂ Emissions, Urbanization to Renewable Energy consumption and GDP Growth were studied. The aim of this study was to provide statistical evidence for Global Warming, using CO₂ emissions as a surrogate and suggest some strategies for the reduction in CO₂ emissions. The conclusions drawn from the analysis were used to suggest the strategies for the reduction in the global warming.

The work was done using Time Series Modelling and Forecasting techniques such as ARIMA Models, Seasonal ARIMA Models. Future projections have been made, which is showing about 1°C increase in global temperature in next two decades. Further, nation wise analysis of CO₂ emissions has been done and clear inequality between developed and developing nations is established. Regression Analysis and Random Forests were used to analyze the various factors affecting the CO₂ Emissions. Renewable Energy and Urbanization are turned out to be important factors. Time Series Regression was also used to include the trend and seasonality components.

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Outline of the Work:

- Data Imputation: Dealing with the missing values
- Simple Exploratory Analysis: Visualizing the data
- Global Warming: Model fitting and Forecast
- Global Warming: Causes and Impacts
- CO₂ Emissions: Model fitting and Forecast
- Regression Analysis Random Forest: Understanding factors related to CO₂ Emissions
- Time Series Analysis: Which countries fare similarly when it come to CO₂ emissions

Software and Tools used:

- R Studio Version 1.2.1335
- Minitab 17
- Gapminder Tools Offline v.4.1.0
- MS Excel 2017

INTRODUCTION

The issues involved in understanding global warming and taking actions to slow its harmful impacts are major environmental challenge of the modern era. It poses a unique blend of challenges with a lot of scientific and economic uncertainties and will cast a shadow over the globe for decades, perhaps even centuries, to come. A careful look at the issues reveals that there is at present no definite answer as to how quickly nations should move to slow climate change. The scientific basis of global warming is widely discussed from different research papers across the disciplines^[1].

Although the exact forecast of extent of warming is highly uncertain, recent evidence and model predictions suggest that global mean surface temperature will rise sharply in the next century and beyond. The fourth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC 2007), gives a good estimate of the global temperature increase over the coming century as from 1.8 to 4°C. This change is much more rapid than any changes that have occurred in the past 10 millennia^[2]. In its 2013 fifth assessment report published in 2013, the IPCC stated in its summary for policymakers that it is “extremely likely that more than half of the observed increase in global average surface temperature” from 1951 to 2010 was caused by human activity.

Despite global efforts to address climate change, including the landmark 2015 Paris climate agreement, carbon dioxide emissions from fossil fuels continue to rise, hitting record levels in 2018. Countries that signed the UN Framework Convention on Climate Change adopted a target to stop the average global temperature from rising before it reaches 2°C above pre-industrial levels.

How have past climate models fared?

Scientists have been making projections of future global warming using climate models of increasing complexity for the past 40-42 years. “Carbon Brief”^[3] has collected prominent climate model projections since 1973 to see how well they project both past and future global temperatures. A short summary of each model is given below.

[1] : A Question of Balance, William Nordhaus

[2] : IPCC report 2013

[3] : www.carbonbrief.org

1) Sawyer (1973)

One of the first projections of future warming came from John Sawyer at the UK's Met Office in 1973. In a paper published in *Nature* in 1973, he hypothesised that the world would warm 0.6°C between 1969 and 2000, and that atmospheric CO₂ would increase by 25%. Sawyer did not provide an estimated warming for each year, just an expected 2000 value. His warming estimate of 0.6C was nearly spot on, however, he overestimated the year 2000's atmospheric CO₂ concentrations.

2) Broecker (1975)

The first available projection of future temperatures due to global warming appeared in an article in *Science* in 1975 published by Columbia University scientist Prof Wally Broecker. He used a Simple Energy Balance Model to estimate what would happen to the Earth's temperature if atmospheric CO₂ continued to increase rapidly after 1975. Broecker's projected warming was reasonably close to observations for few decades, but recently has been considerably higher. This is again mostly due to overestimation of CO₂ concentrations.

3) Hansen et al (1981)

NASA's Dr James Hansen and colleagues published a paper in 1981 that also used a simple energy balance model to project future warming, but accounted for thermal inertia due to ocean heat uptake. Hansen and colleagues presented a number of different scenarios, varying future emissions and climate sensitivity. The overall rate of warming between 1970 and 2016 projected by Hansen et al in 1981 in the fast-growth scenario has been about 20% lower than observations.

IPCC has projected future warming in its every assessment report on Climate Change (1990, 1995, 2001, 2007 and 2013). Below is a summary of all the models with the difference in the rate of warming between each model and NASA's temperature observations.

Model	Difference in 1970-2016 mean warming rate vs. Obs
Broecker 1975	+30%
Hansen et al 1981	-20%
Hansen et al 1988	+30%
IPCC 1st Report, 1990	+17%
IPCC 2nd Report, 1995	-28%*
IPCC 3rd Report, 2001	-14%
IPCC 4th Report, 2007	+8%
IPCC 5th Report, 2013	+16% (+9%)#

* Differences are calculated over 1990-2016, as estimates prior to 1990 are not available.

Difference in the parenthesis are based on blended models

Models are far from perfect and will continue to be improved over time. They also show a fairly large range of future warming that cannot easily be narrowed using just the changes in climate that we have observed. Nevertheless, the close match between projected and observed warming since 1970 suggests that estimates of future warming may prove similarly accurate.

Data Description

1. Data 1: Data about per capita CO₂ emissions (in tonnes). Data on CO₂ emissions from 1902 to 2006 of 36 countries. (Source – World Bank)
2. Data 2: Data about per capita Energy consumption (tonnes of oil equivalent). Data from 1971 to 2010 of 108 countries. (Source-World Bank)
3. Data 3: Data on average global sea level from 1993 to 2015 (Source-NASA).
4. Data 4: Data from 1960 to 2018 on population of all the countries. (Source-World Bank)
5. Data 5: Data from 1961 to 2006 on GDP growth rate of 26 major economies in the world. (Source-World Bank)
6. Data 6: Data from 1991 to 2006 on renewable energy consumption (% of total energy consumption) of 34 countries. (Source-World Bank)
7. Data 7: Data from 1961 to 2006 on urban population (% of total population) of 34 countries. (Source-World Bank)
8. Data 8: Data from 1994 to 2015 on forest planted (in hectares) of 34 countries. (Source-World Bank)
9. Data 9: Global land average temperature from 1750 to 2015. (Source-Kaggle)
10. Data 10: Global mean temperature change (1811-2018) from pre-industrial average. (Source-NASA)

Handling Missing Data

The problem of missing data was handled using imputation technique called Multivariate Imputation by Chained Equations (MICE). MICE gives multiple imputations (replacement values) for multivariate missing data. The method is based on Fully Conditional Specification, where each incomplete variable is imputed by a separate model.

The method of imputation used in this case is Classification and Regression Trees (CART).

The procedure followed is as follows (R package MICE):

1. Fit a classification or regression tree by recursive partitioning;
2. For each missing value, find the terminal node they end up according to the fitted tree;
3. Make a random draw among the members in the node, and take the observed value from that draw as the imputation.

The missing data is as follows:

Year	Urbanisation %	Co2 emissions per capita	Agricultural land	Methane Emissions	GDP growth per capita	Population Growth %
2010	30.930	1.3810349799999999	60.397418260000002	621480	7.0423490319999997	1.350339
2011	31.276	1.456958873	60.430043150000003	628938.06359999999	3.8939002170000001	1.288576
2012	31.634	1.5705901710000001	60.420625659999999	636395.82720000006	4.1655280819999998	1.231623
2013	32.003	1.562597867	60.439460650000001	NA	5.1349569070000003	1.183016
2014	32.384	1.706162256	60.44719645	NA	6.1867319829999996	1.145624
2015	32.777	1.738971219	60.44719645	NA	6.7970397499999997	1.116633
2016	33.182	1.7954231199999999	60.44719645	NA	6.9970095040000002	1.089880
2017	33.600	1.841996698	NA	NA	6.0351545279999996	1.062597
2018	34.030	NA	NA	NA	5.8783168300000002	1.037323

And the imputed output is:

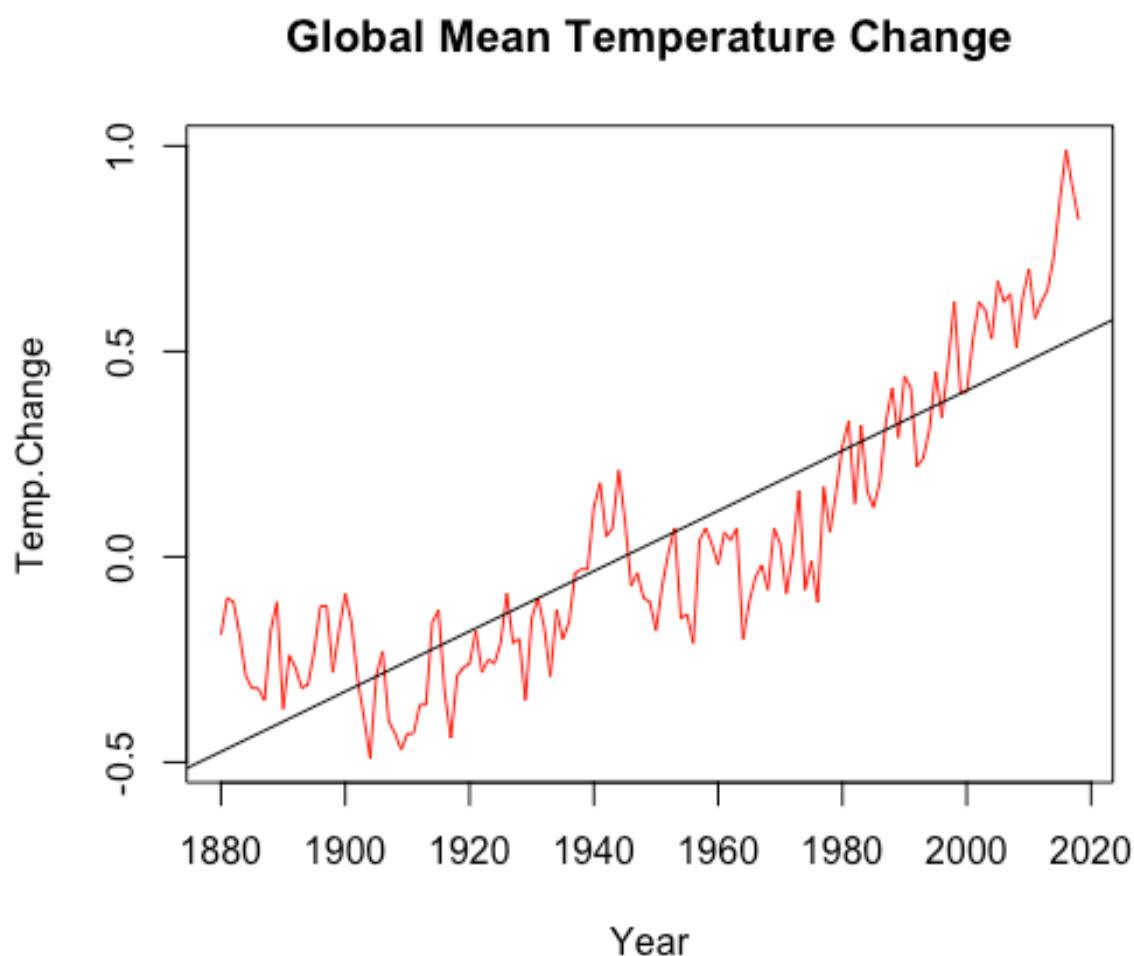
Urbanisation..	Co2.emissions.per.capita	Agricultural.land	Methane.Emissions	GDP.growth.per.capita	Population.Growth..
30.930	1.3810350	60.39742	621480.0	7.0423490	1.350339
31.276	1.4569589	60.43004	628938.1	3.8939002	1.288576
31.634	1.5705902	60.42063	636395.8	4.1655281	1.231623
32.003	1.5625979	60.43946	636395.8	5.1349569	1.183016
32.384	1.7061623	60.44720	636395.8	6.1867320	1.145624
32.777	1.7389712	60.44720	628938.1	6.7970397	1.116633
33.182	1.7954231	60.44720	606364.0	6.9970095	1.089880
33.600	1.8419967	60.56391	628938.1	6.0351545	1.062597
34.030	1.4569589	60.44720	621480.0	5.8783168	1.037323

Source : <https://www.rdocumentation.org/packages/mice/versions/3.6.0/topics/mice.impute.cart>

Simple Exploratory Analysis

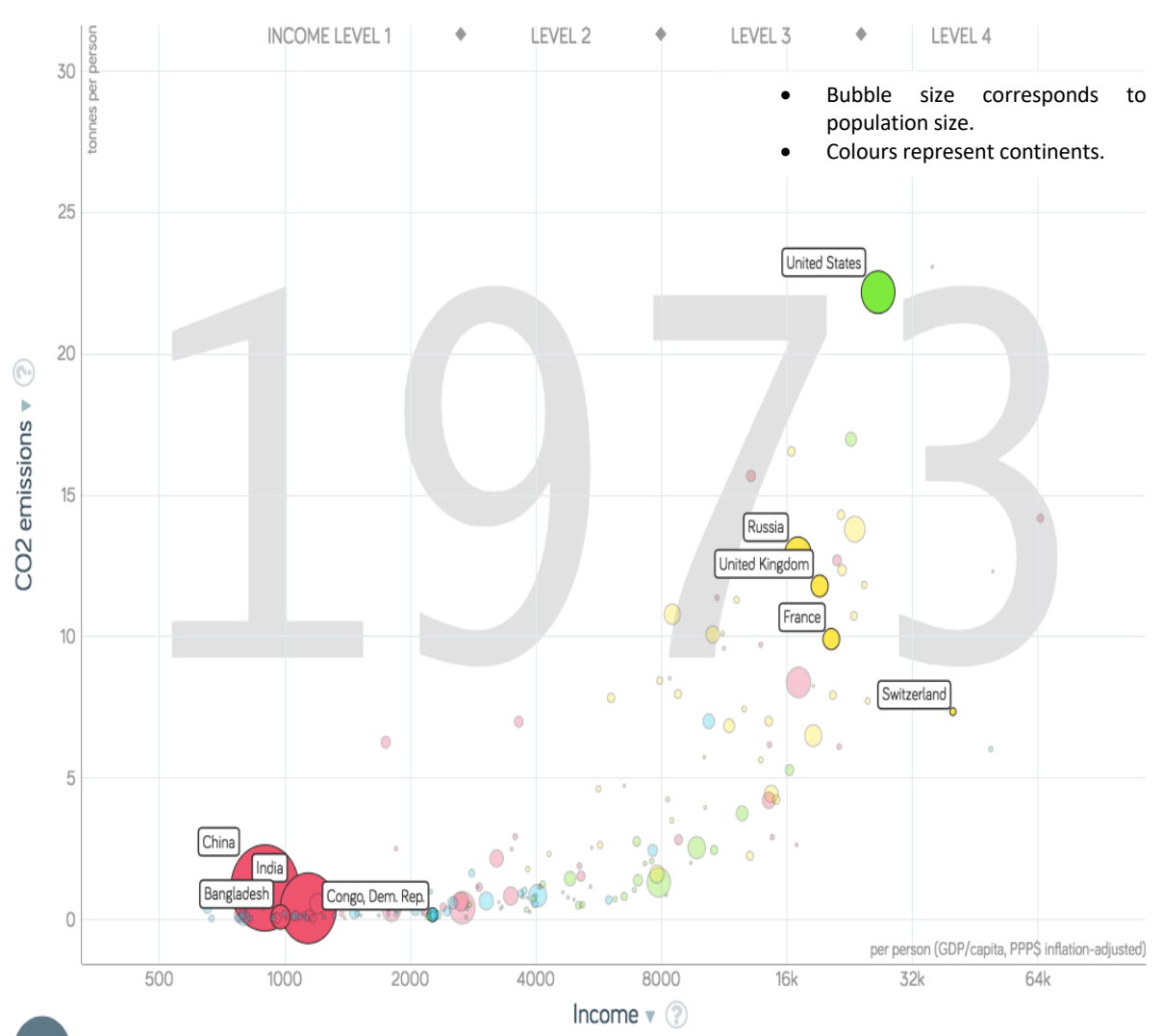
1) Global Temperature Data

This data contain mean temperature change for every year from pre-industrial period average. Data is available from 1880 to 2018.

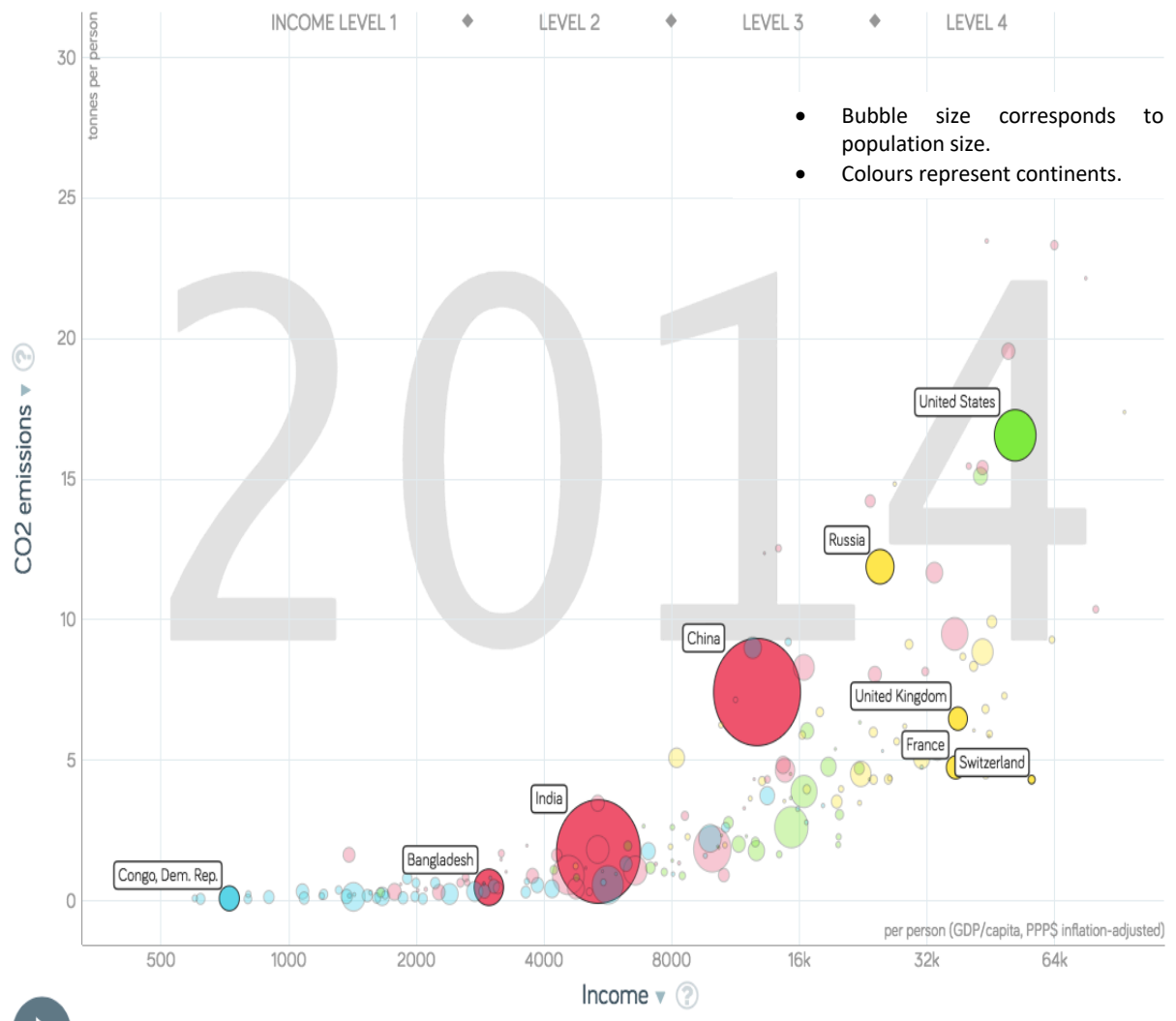


We see that over the last few decades, temperatures have risen sharply at the global level — to approximately 0.8°C higher than 1960-1980 baseline. When extended back to 1850, we see that temperatures then were a further 0.4°C colder than they were in our 1960-1980 baseline. Overall, if we look at the total temperature increase since pre-industrial times, this therefore amounts to approximately 1.2°C . We have now surpassed the 1°C mark, an important marker as it brings us more than halfway to the global limit of keeping warming below 2°C .

2) CO₂ Emissions and Prosperity

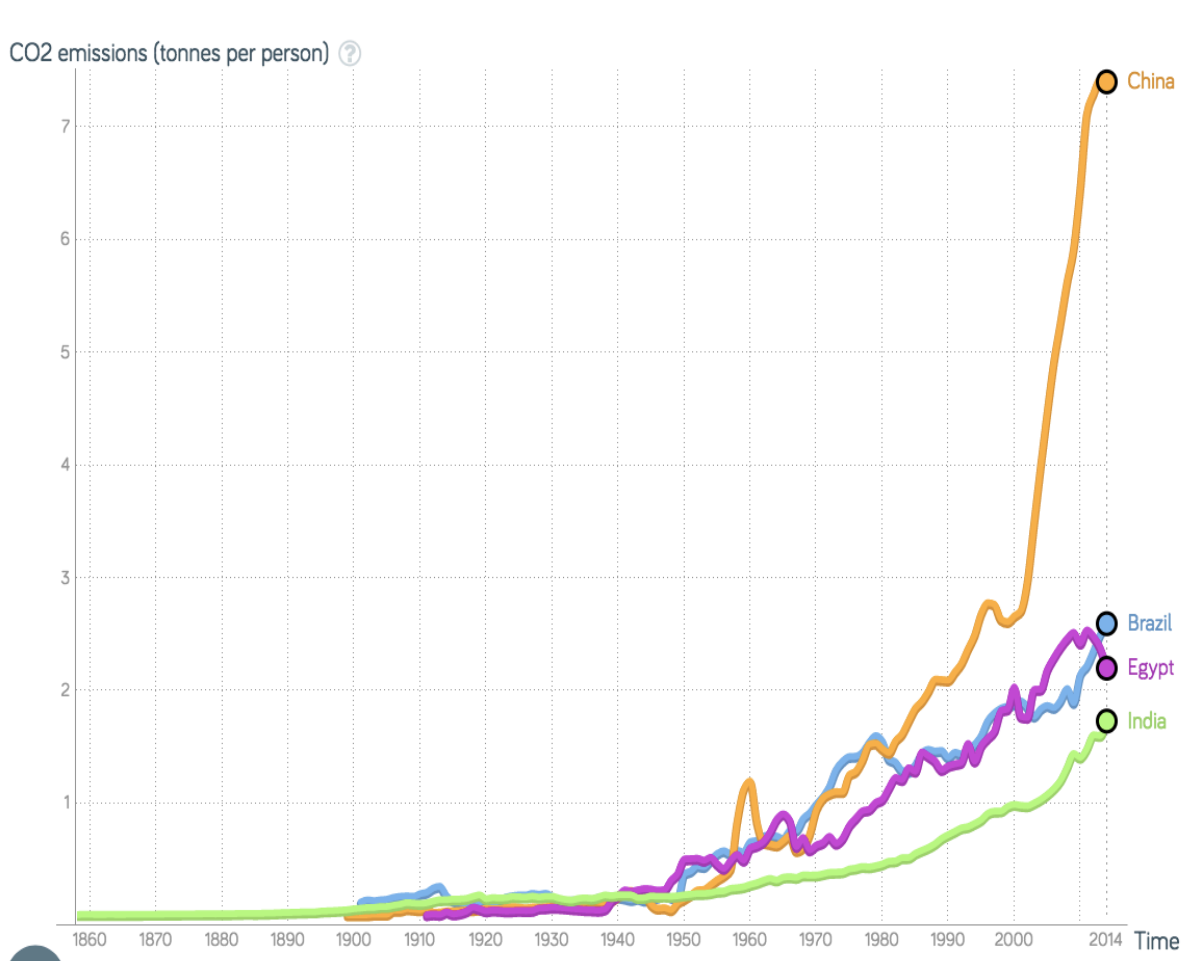


Historically, CO₂ emissions have been primarily driven by increasing fuel consumption. Energy consumption has been, and continues to be, a fundamental pillar of economic growth and poverty alleviation. As a result, we see that there is strong correlation between per capita CO₂ emissions and GDP per capita.



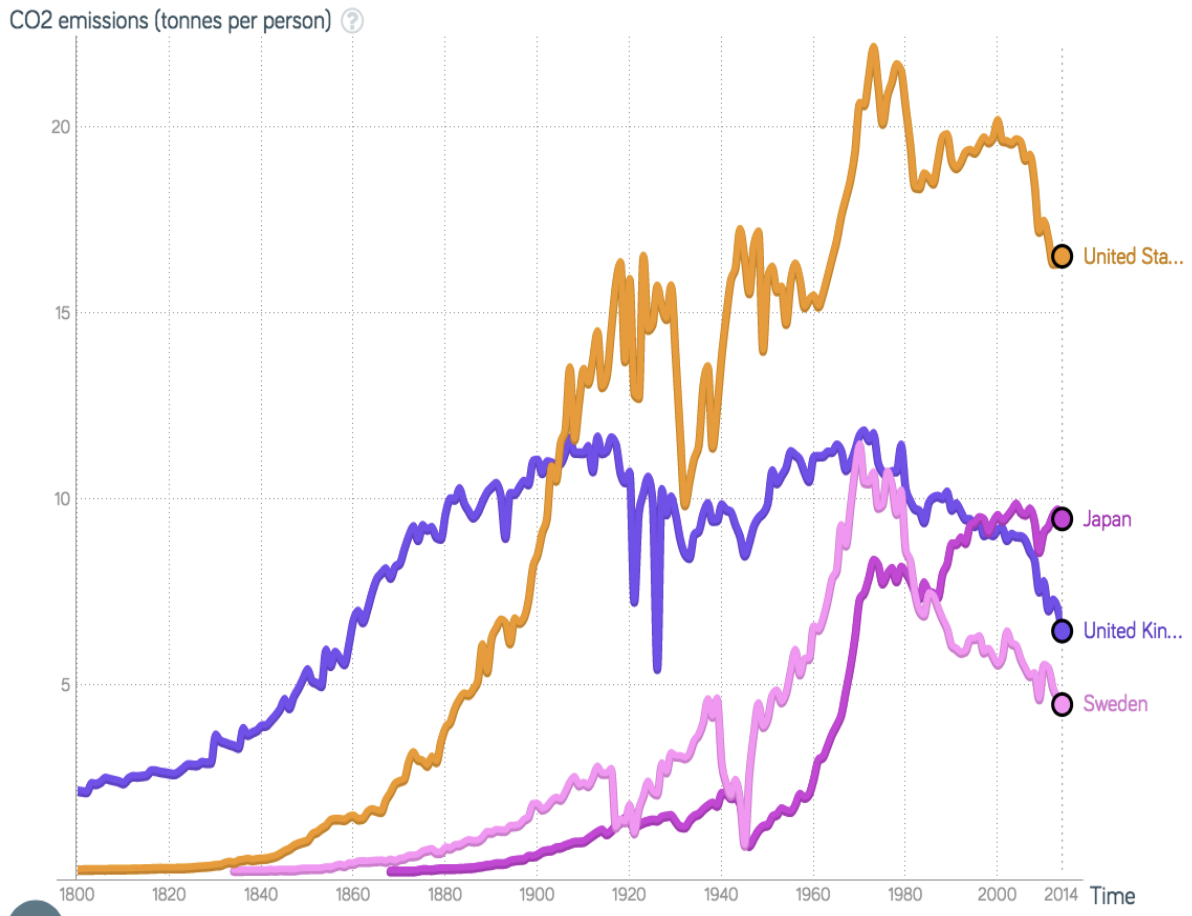
The correlation is also present over time: Most countries begin in the bottom left of the chart at low CO₂ and low GDP, and move upwards and to the right. Historically, where fossil fuels are the dominant form of energy, we therefore see increased CO₂ emissions as an unintended consequence of development and economic prosperity. While we see this general relationship between CO₂ and GDP, there are outliers (Sweden, Switzerland) in this correlation, and important differences exist in the rate with which per capita emissions have been growing.

4) Per Capita CO₂ Emissions of Developing Nations



All of these countries show increasing trend in their emissions. For China, there is huge increase from 2001. Primary reason for that is China became a member of World Trade Organization (WTO) in Dec 2001, which contributed to the growth of the Chinese Economy and consequently CO₂ emissions started increasing.

5) Per Capita CO₂ Emissions of Developed Nations



All these countries had increasing trend in CO₂ emissions till the end of last century. After that, UK, USA and Sweden have reduced their CO₂ emissions significantly. USA's emissions are still at very high level compared to other nations.

Analysis of CO₂ emissions in India

The data set contains the CO₂ emissions of India measured for the years 1960-2018. The regressors include Urbanization (%), Renewable Energy (%), Agricultural Land (%), Methane Emissions (Tonnes equivalent to CO₂), GDP Growth per capita and Population Growth (%).

The data for Renewable Energy is available from 1990. Hence for the first part of the analysis, data from the years 1990-2018 is considered.

1. Multiple Linear Regression

Multiple linear regression model was fit on the training data, which is 80% of the total observations. The summary is as follows:

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -4.048e+00  7.172e+00  -0.564  0.58378
Urbanisation..  7.370e-02  6.599e-02   1.117  0.28788
Renewable.energy.. -3.472e-02  6.150e-03  -5.645  0.00015 ***
Agricultural.land  3.537e-02  1.001e-01   0.353  0.73054
Methane.Emissions  2.248e-06  1.478e-06   1.521  0.15641
GDP.growth.per.capita -3.174e-03  3.153e-03  -1.006  0.33583
Population.Growth..  7.676e-01  4.798e-01   1.600  0.13795
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02257 on 11 degrees of freedom
Multiple R-squared:  0.9939,    Adjusted R-squared:  0.9905
F-statistic: 297.1 on 6 and 11 DF,  p-value: 1.638e-11
```

The Adjusted R² value is 0.99, is very close to 1. Hence the model explains the variation in the response well. Renewable energy is the only variable that is significant.

The correlation matrix shows variables are highly correlated. This poses the problem of multicollinearity. A method for detecting multicollinearity is to observe the VIFs, that is the Variance importance factors. The maximum VIF for this dataset is greater than 400 which denote severe multicollinearity. Thus, the model obtained is unstable.

To overcome this problem of multicollinearity, Principal Component Regression is used.

2. Principal Component Regression

The method of Principal Component Regression uses the Principal Components (PCs) as the regressors. The PCs are linear combinations of the regressors and are orthogonal to one another. Thus, the transformed regressors are now uncorrelated to each other.

The summary is as follows:

```
Cross-validated using 10 random segments.
      (Intercept) 1 comps 2 comps 3 comps 4 comps 5 comps 6 comps
cv          0.2489 0.06899 0.05631 0.03920 0.03414 0.03780 0.04184
adjcv       0.2489 0.06873 0.05593 0.03888 0.03369 0.03712 0.04094

TRAINING: % variance explained
              1 comps 2 comps 3 comps 4 comps 5 comps 6 comps
X              84.54  98.12  99.60  99.87  99.98 100.00
Co2.emissions.per.capita 92.91  95.73  98.07  98.82  98.83  98.84
```

The first PC explains 92.91 % of variation in response, hence used as regressor. The coefficients of the PC regression model are as follows:

```
              Co2.emissions.per.capita
Urbanisation.. 0.04592644
Renewable.energy.. -0.04528906
Agricultural.land -0.04470006
Methane.Emissions 0.04583914
GDP.growth.per.capita 0.02230418
Population.Growth.. -0.04584416
```

3. Simple Linear Regression

The only significant regressor in multiple linear regression model is Renewable Energy. A simple linear model was fit to this variable to observe the effect of Renewable Energy only.

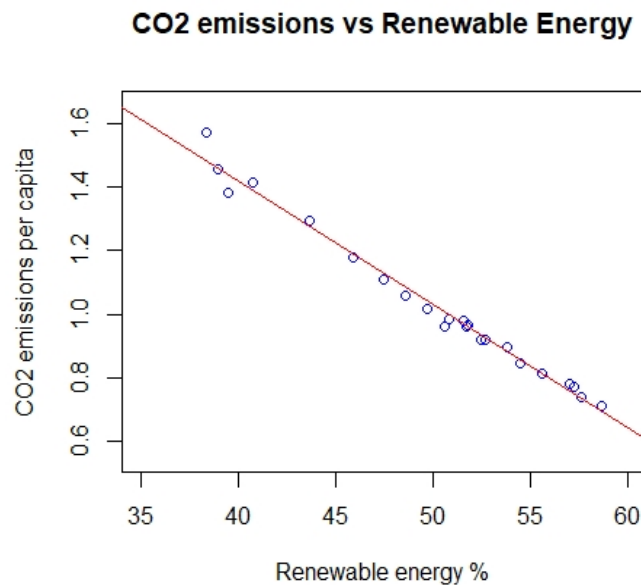
The output from R is as follows:

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.969766   0.062928   47.19 < 2e-16 ***
Renewable.energy.. -0.038802   0.001255  -30.91 1.07e-15 ***
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03255 on 16 degrees of freedom
Multiple R-squared:  0.9835,    Adjusted R-squared:  0.9825
F-statistic: 955.5 on 1 and 16 DF,  p-value: 1.071e-15
```

We see that even if we use renewable energy as the only regressor, Adjusted R^2 is still 0.9825 which is really good. Thus, renewable energy explains variation in CO₂ emissions really well.

The scatter plot is as below:



We can see clear decreasing linear relationship between CO₂ emissions and Renewable Energy. From the model, we can conclude that if we want to restrict our CO₂ emissions per capita under 1.5 tonnes, we need to use at least 40 % of Renewable energy in our total energy consumption.

To study the effect of the other variables not including Renewable Energy, we have done analysis on rest of the variables, for years 1960 – 2018.

1. Multiple Linear Regression

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.581e-01	3.475e+00	0.276	0.784130
Urbanisation..	6.199e-02	1.457e-02	4.255	0.000118 ***
Agricultural.land	-8.656e-03	6.806e-02	-0.127	0.899424
Methane.Emissions	-4.025e-07	6.429e-07	-0.626	0.534748
GDP.growth.per.capita	-8.615e-04	3.740e-03	-0.230	0.818990
Population.Growth..	-5.309e-01	1.494e-01	-3.554	0.000972 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

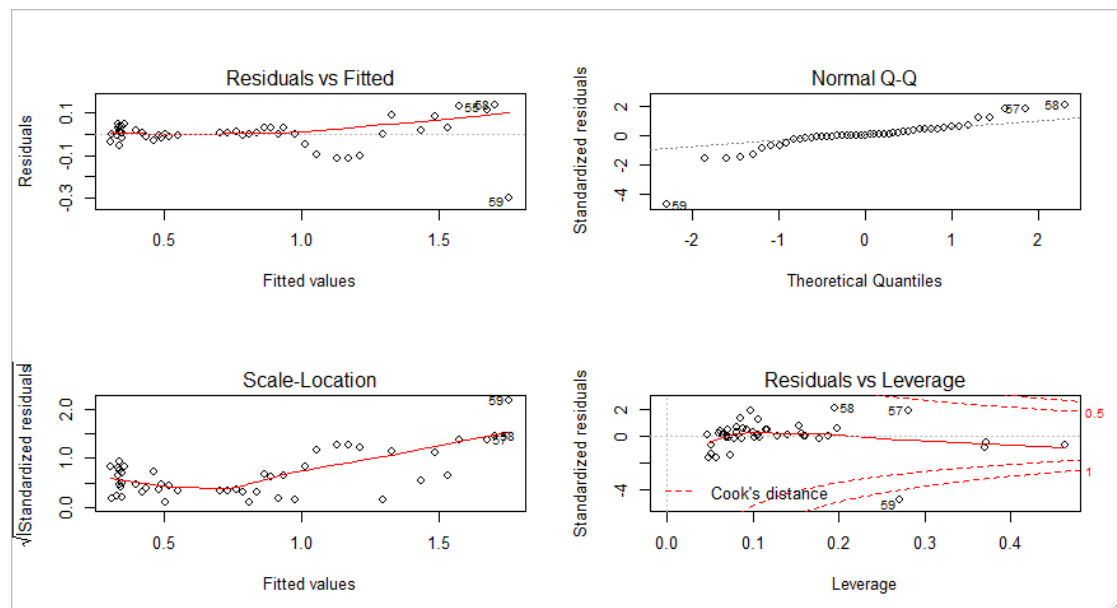
Residual standard error: 0.06962 on 41 degrees of freedom

Multiple R-squared: 0.9792, Adjusted R-squared: 0.9767

F-statistic: 386.5 on 5 and 41 DF, p-value: < 2.2e-16

This model has Urbanisation and Population Growth as significant variables. Adjusted R² is 0.977 which is very close to 1.

Residual analysis as follows:

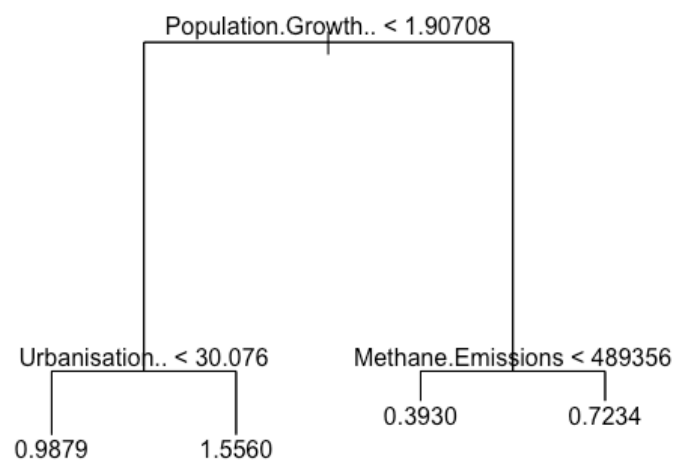


The first plot shows some pattern in the residuals. Q-Q plot shows departure from normality. Hence, we can say this model is not a good model.

2. Regression Trees and Random Forest

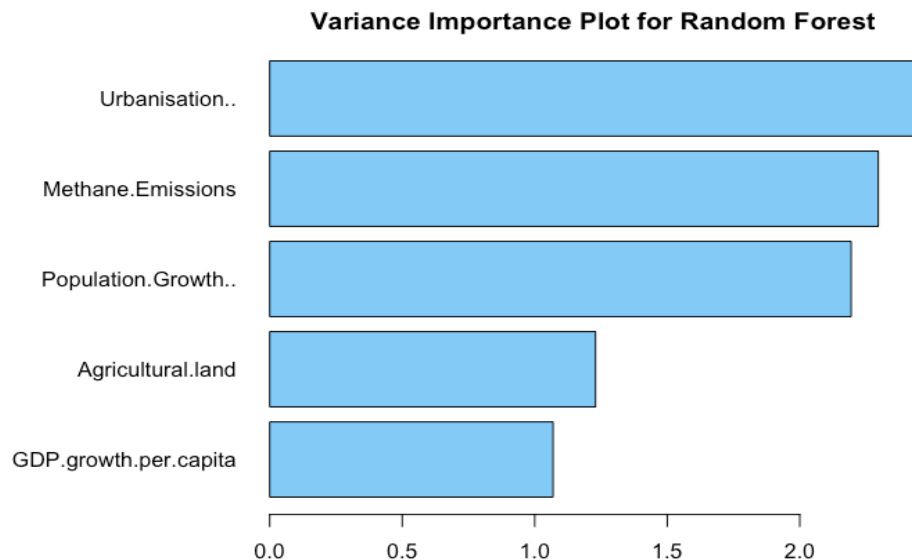
Regression Tree is a method which recursively splits data into nodes. A series of decision lands a new observation in the last node, called as leaf node and prediction is based on the observations present in that node. Trees are very easy to interpret but are not robust. Hence Random Forests method is used, which makes use of a large number of trees to get prediction on response.

An example of a tree fit on this data is as follows:



This tree is very easy to interpret. Increase in Methane emissions and urban population corresponds to higher CO2 emissions. Trees are not robust, hence model is not stable.

Random Forest was carried out on the same data. A Variable Importance plot gives us an idea about significance of contribution of each variable in the model.



Here Urbanization, Methane Emissions and Population Growth are the important variables. Random forest gives a robust model compared to trees.

4. K – Nearest Neighbour Regression

This method finds out K most similar observation points to the desired data point and gives prediction for response based on all of them. Here mean of responses of all K neighbours is used.

K was decided by finding out for which value the test error is minimum. K = 6 was found out to be maximum. The distance metric used was Euclidean distance.

Comparison of all models:

1. Using all regressors

Estimates of test error for all models is as shown on the next page:

Model	Description	Test Error
Multiple Linear Regression	Simple Additive Model with all the Regressors	0.0030
Simple Regression	Significant variable from above model	0.0027
Principal Component Regression	First PC considered as regressors	0.0050

The predictions on test set along with actual test set responses are :

Actual	Multiple Regression	Simple Regression	Principal Component Regression
0.7712369	0.7403507	0.7505318	0.7472251
0.7823041	0.7539934	0.7599165	0.7533925
0.8111213	0.8037086	0.8140577	0.8090130
0.9612670	0.9952630	1.0034058	0.9983211
1.5705902	1.4613496	1.4657357	1.4294147

Simple Regression model gives the minimum test error.

2. Using all regressors except Renewable Energy

The test error for all models is as follows:

Model	Description	Test Error
Multiple Linear Regression	Simple Additive Model with all the Regressors	0.0049
Regression Trees	Identification of important variables	0.0187
Random Forest	Identification of important variables, but more stable than Trees	0.0090
K Nearest Neighbour Regression	Predictions based on most similar observations	0.0106

The predictions on test set along with actual test set responses are:

Actual	Multiple Regression	Trees	RandomForest	KNN
0.3770322	0.3171657	0.3940816	0.3756617	0.3452548
0.3808329	0.3346111	0.3940816	0.4215009	0.3540111
0.2682564	0.3384588	0.3940816	0.4317681	0.3625560
0.4230476	0.4281782	0.3940816	0.4646193	0.3815515
0.5694158	0.5744785	0.3940816	0.5679047	0.5808243

The test error in case of Multiple Regression is minimum.

The following data set contains the CO₂ emissions of India measured for the years 1990-2014. The regressors include Agricultural land (%), Forest cover

(%), Sulphur emissions per person, Energy per person in kg of oil equivalent, GDP growth per capita in USD, NO₂ emissions in thousand metric tonnes CO₂ equivalent, Population, Annual rainfall in mm.

1. Multiple Linear Regression

Multiple linear regression model was fit on the training data, which is 80% of the total observations. The glm() command in R was used. The summary is as follows:

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -3.807e+00  7.157e+00  -0.532  0.60639
Agricultural.Land.. 6.596e-02  1.050e-01   0.628  0.54393
Forest.Cover.. -5.733e-02  1.427e-01  -0.402  0.69633
Planted.forest.in.hect -2.276e-03  1.469e-03  -1.548  0.15254
sulphur.emissions.per.person 7.951e-02  5.490e-02   1.448  0.17814
Energy.per.person.in.kg.of.oil.equivalent 2.723e-03  7.043e-04   3.866  0.00313 **
GDP.growth.per.capita.in.USD -3.401e-03  4.787e-03  -0.710  0.49364
NO2.emissions.in.thousand.metric.tonnes.Co2.equivalent 6.958e-07  1.820e-06   0.382  0.71020
Population 5.629e-10  7.113e-10   0.791  0.44708
Annual.rainfall.in.mm -1.303e-05  1.173e-04  -0.111  0.91371
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02501 on 10 degrees of freedom
Multiple R-squared:  0.9949,    Adjusted R-squared:  0.9903
F-statistic: 217.1 on 9 and 10 DF,  p-value: 2.881e-10

```

The regressor Energy consumption is significant. Residual analysis shows model is fairly good. The maximum of the VIFs is greater than 100. Thus, multicollinearity is detected.

2. Stepwise Regression

Stepwise Regression aims to reduce the number of regressors. This addresses the problem of multicollinearity. The only variables that are important in explaining variation in response are retained in the model. The model was fit to the data using step() command in R. The summary is as follows:

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.201e+00  9.139e-01   1.314  0.2116
Forest.Cover.. -1.060e-01  5.404e-02  -1.962  0.0716 .
Planted.forest.in.hect -2.381e-03  1.184e-03  -2.012  0.0655 .
sulphur.emissions.per.person 6.701e-02  4.014e-02   1.669  0.1189
Energy.per.person.in.kg.of.oil.equivalent 3.008e-03  3.347e-04   8.989 6.11e-07 ***
GDP.growth.per.capita.in.USD -4.255e-03  2.840e-03  -1.498  0.1581
Population 6.893e-10  2.784e-10   2.476  0.0278 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

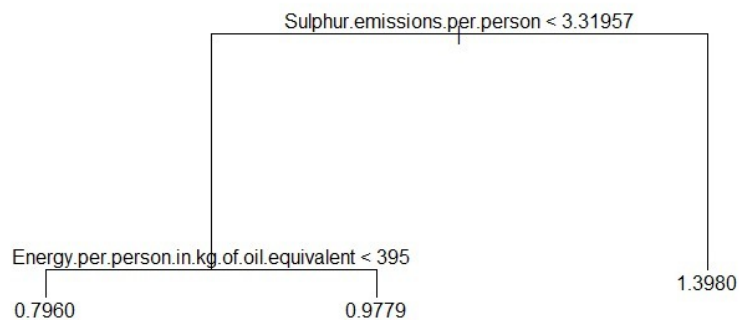
Residual standard error: 0.0225 on 13 degrees of freedom
Multiple R-squared:  0.9946,    Adjusted R-squared:  0.9922
F-statistic: 402.2 on 6 and 13 DF,  p-value: 5.466e-14

```

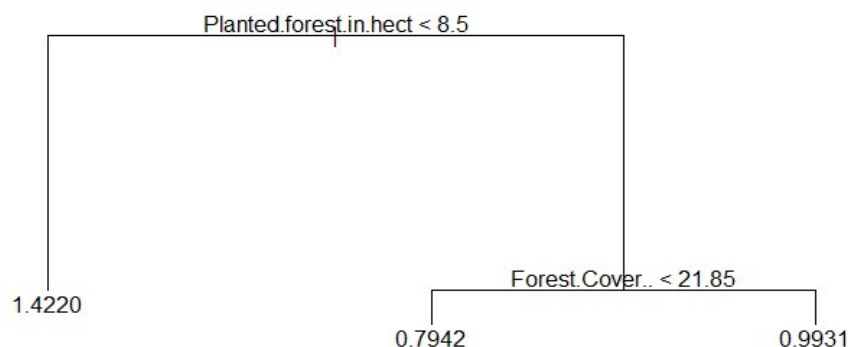
Stepwise regression shows reduction in number of variables. Energy per person, Population are significant variables. The residual analysis shows model is fairly stable.

3. Regression Trees and Random Forest

A couple of trees fit on the data are as follows:

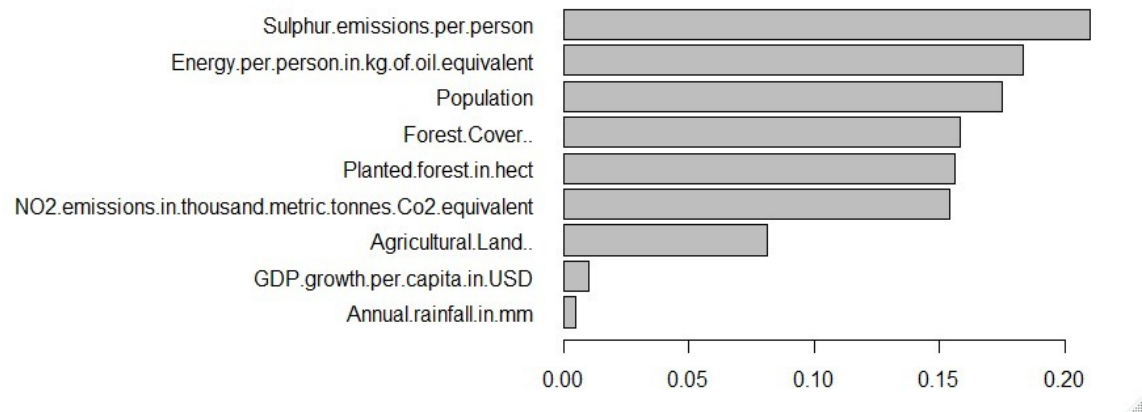


Increase in Sulphur emissions and Energy per person corresponds to higher CO₂ emissions. Trees are not robust, hence model is not stable. Another tree fit on the data is as given below:



Increase in Forest Cover corresponds to increase in CO₂. This may indicate the adverse effect of increasing CO₂ emissions on Forest covers and problems like Forest fires. There are theories in the scientific community regarding higher CO₂ corresponding to drier forest atmosphere.

The Variable Importance plot for the data using Random Forest model as follows:



Sulphur Emission, Energy per person and Population are the important variables. Random forest gives a robust model.

Performance on Test data:

1. Test error for all model

Model	Description	Test Error
Multiple Linear Regression	Simple Additive Model with all the Regressors	0.0005
Stepwise Regression	Reduction in number of variables in a stepwise manner	0.0003
Regression Trees	Identification of important variables	0.1215
Random Forest	Identification of important variables, but more stable than Trees	0.0198

2. Predictions for test data and actual responses in test set

Actual	Multiple Linear	Tree	Random Forest	Stepwise Regression
0.712	0.6971952	0.801800	0.7839919	0.7095587
0.845	0.8524219	0.801800	0.8803962	0.8494919
1.070	1.0629929	0.985000	1.0590706	1.0557967
1.590	1.6087633	1.401667	1.4680658	1.6146391
1.730	1.6902408	0.985000	1.4514739	1.7010372

Time Series Regression

Time Series Regression assumes the fact that a time series y has a linear relation with other regressor time series x .

The `tslm()` function from package `forecast` in R fits a linear regression model to time series data. It is similar to the `lm()` function but provides additional facilities for handling time series.

The first data set contains the CO₂ emissions of India measured for the years 1990-2012. The regressors include Urbanization (%), Renewable Energy (%), Agricultural Land (%), Methane Emissions (Tonnes equivalent to CO₂), GDP Growth per capita and Population Growth (%).

A multiple time series regression model is fit. The summary is as follows :

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -2.730e+00  9.305e+00  -0.293  0.772984
Urbanisation.. -1.641e-02  7.925e-02  -0.207  0.838605
Renewable.energy.. -3.762e-02  7.877e-03  -4.776  0.000206 ***
Agricultural.land  8.779e-02  1.210e-01   0.725  0.478633
Methane.Emissions  1.478e-06  1.910e-06   0.774  0.450278
GDP.growth.per.capita -4.003e-03  4.023e-03  -0.995  0.334578
Population.Growth.. -3.114e-02  5.470e-01  -0.057  0.955304
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0308 on 16 degrees of freedom
Multiple R-squared:  0.9641,    Adjusted R-squared:  0.9506
F-statistic: 71.52 on 6 and 16 DF,  p-value: 1.175e-10
```

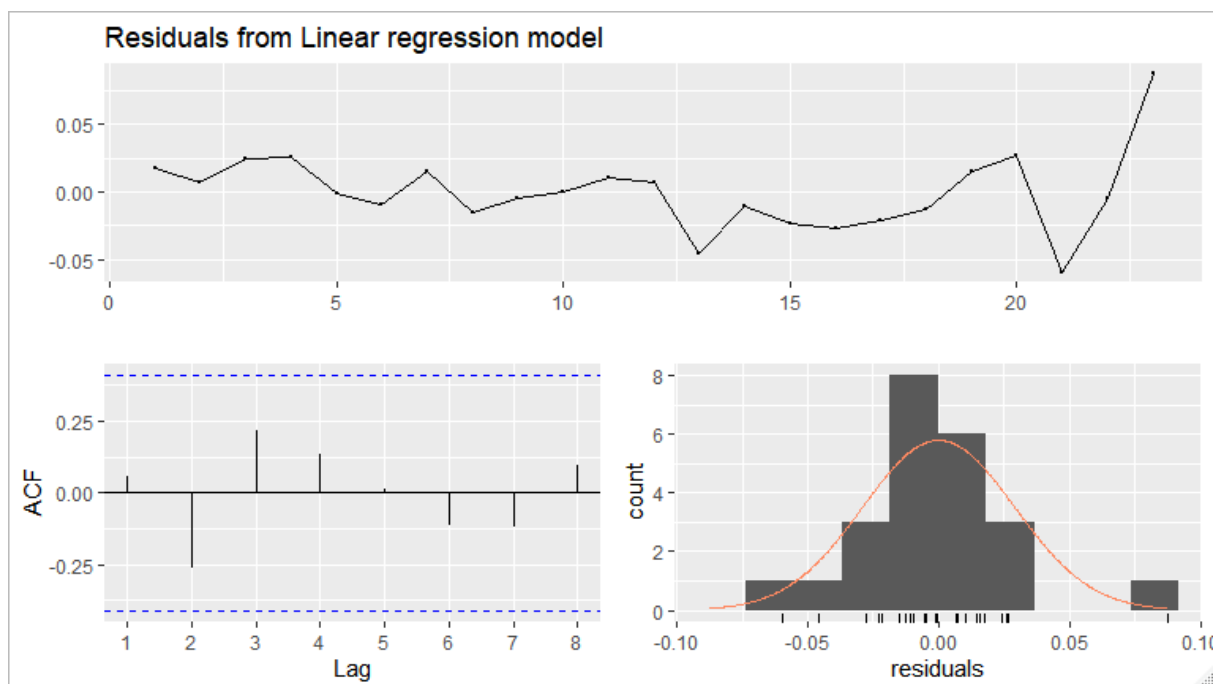
The Adjusted R^2 value is 0.95, is very close to 1. Hence the model explains the variation in the response well. Renewable energy is the only variable that is significant.

A model fit only on Renewable energy is considered. The summary is as follows:

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.983790  0.051613  57.81  <2e-16 ***
Renewable.energy.. -0.039087  0.001026  -38.10  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02976 on 21 degrees of freedom
Multiple R-squared:  0.9563,    Adjusted R-squared:  0.9542
F-statistic: 459.7 on 1 and 21 DF,  p-value: 9.29e-16
```

Residual analysis of this model gives the following plots.



Residuals show no pattern and appear to be random. ACF plot shows all correlations are within bounds.

Breusch-Godfrey test rejects the serial correlation of residuals of order up to 10 with p value 0.2455. Shapiro – Wilks test indicates normality of residuals, with p value 0.1196.

Thus, the time series regression model is fairly good.

The following second data set contains the CO₂ emissions of India measured for the years 1990-2014. The regressors include Agricultural land (%), Forest cover (%), Sulphur emissions per person, Energy per person in kg of oil equivalent, GDP growth per capita in USD, NO₂ emissions in thousand metric tonnes CO₂ equivalent, Population, Annual rainfall in mm.

The summary for a multiple time series regression model is as follows :

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    -2.633e+00  5.801e+00  -0.454  0.6564
Agricultural.Land..  6.144e-02  9.067e-02   0.678  0.5084
Forest.Cover..   -1.058e-01  8.949e-02  -1.183  0.2554
Planted.forest.in.hect  -2.092e-03  1.179e-03  -1.774  0.0964 .
Sulphur.emissions.per.person  7.767e-02  4.491e-02   1.729  0.1042
Energy.per.person.in.kg.of.oil.equivalent  3.087e-03  3.788e-04  8.150 6.84e-07 ***
GDP.growth.per.capita.in.USD  -2.674e-03  3.649e-03  -0.733  0.4750
NO2.emissions.in.thousand.metric.tonnes.Co2.equivalent -1.939e-07  1.173e-06  -0.165  0.8709
Population       7.464e-10  5.363e-10   1.392  0.1843
Annual.rainfall.in.mm  -7.533e-07  8.350e-05  -0.009  0.9929
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02286 on 15 degrees of freedom
Multiple R-squared:  0.9963,    Adjusted R-squared:  0.994
F-statistic: 445.9 on 9 and 15 DF,  p-value: < 2.2e-16

```

The Adjusted R^2 value is 0.99, is very close to 1. Hence the model explains the variation in the response well. Energy consumption per person is the only variable that is significant.

A model fit only on Energy consumption per person is considered. The summary is as follows:

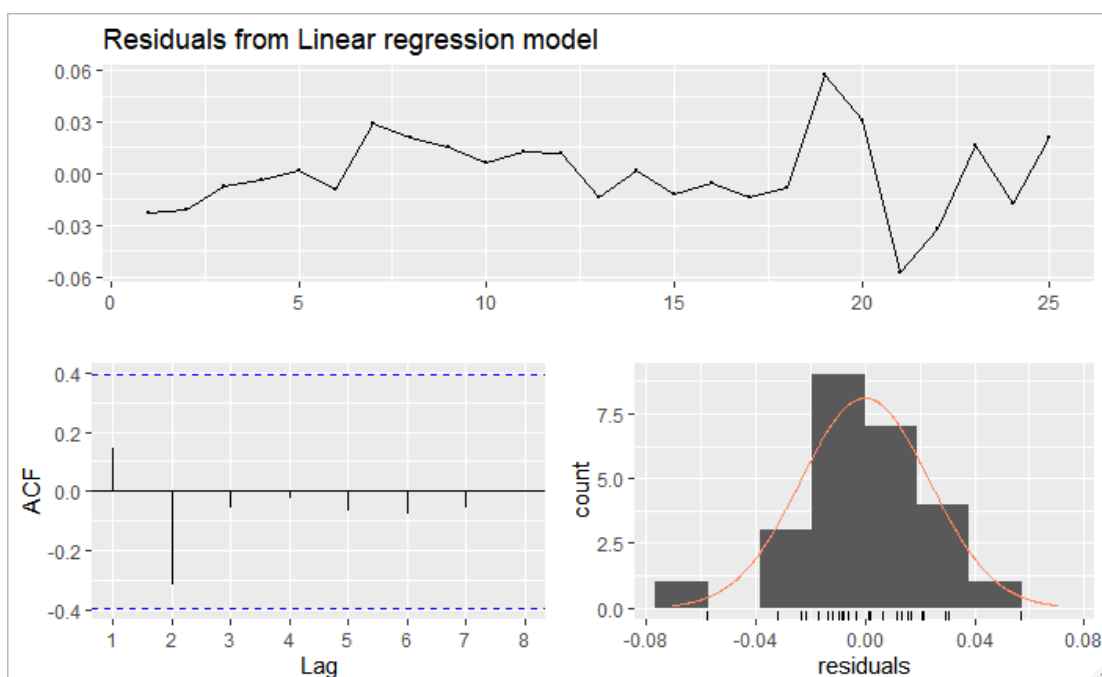
```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    -4.601e-01  2.610e-02  -17.63  7.4e-15 ***
Energy.per.person.in.kg.of.oil.equivalent  3.406e-03  5.639e-05   60.40 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02394 on 23 degrees of freedom
Multiple R-squared:  0.9937,    Adjusted R-squared:  0.9935
F-statistic: 3648 on 1 and 23 DF,  p-value: < 2.2e-16

```

Residual analysis for this model gives the following plots.



Residuals show no pattern and appear to be random. ACF plot shows all correlations are within bounds.

Breusch-Godfrey test rejects serial correlation of residuals of order up to 13 with p value 0.293. Shapiro – Wilks test indicates normality of residuals, with p value 0.8833.

Comparing CO₂ Emissions of Countries according to their income levels

Data: Average of CO₂ Emissions of Countries for years 2000-2014, Income levels as decided by World Bank.

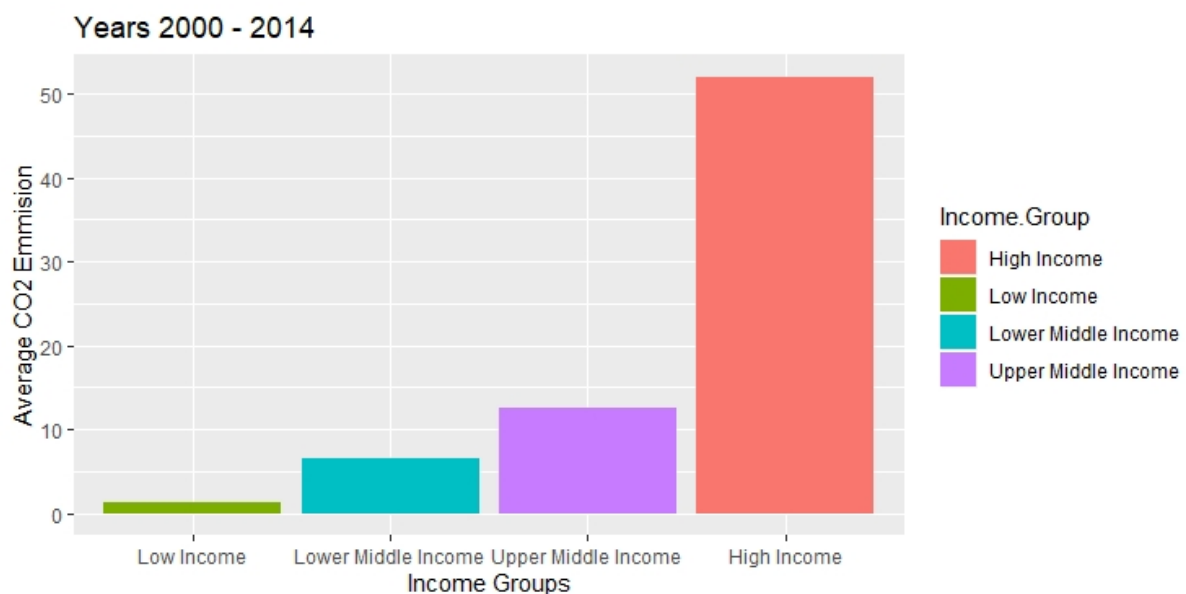
The following analysis was carried out to check if CO₂ emissions of individual groups of countries according to their income are significantly different. The four income groups are as follows – High Income, Upper Middle Income, Lower Middle Income and Low Income.

Exploratory analysis was done to observe group wise behaviour.

Summary of the data as follows:

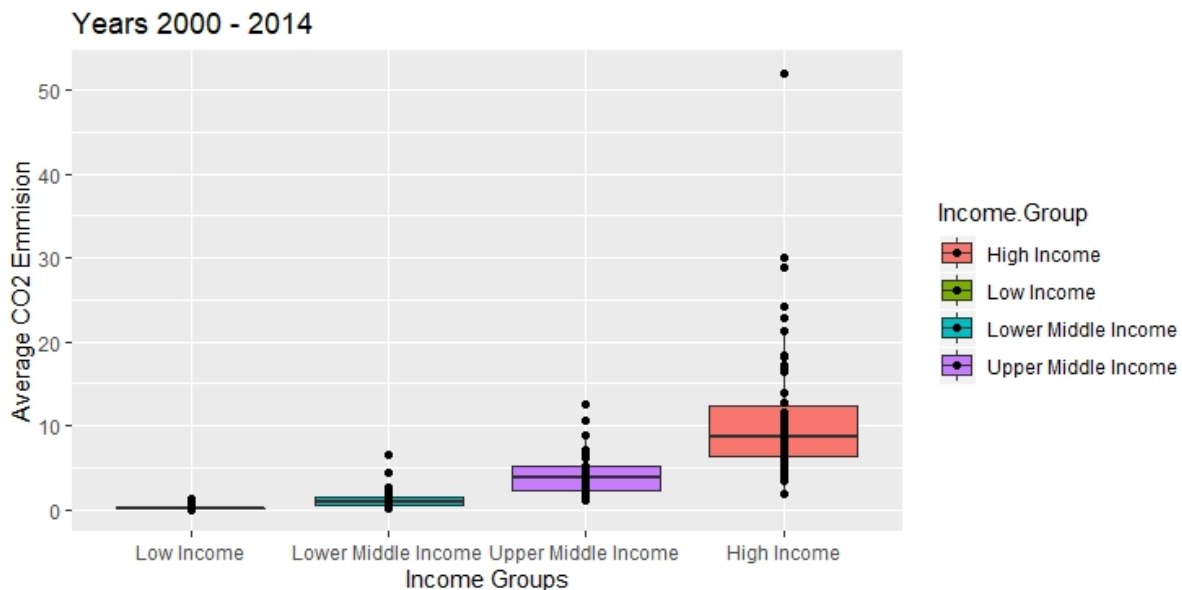
```
> summary(Data)
      Income.Group      Avg
High Income      :50  Min.   : 0.02842
Low Income       :32  1st Qu.: 0.56507
Lower Middle Income:35 Median : 2.64582
Upper Middle Income:38 Mean   : 5.03179
                  3rd Qu.: 6.93390
                  Max.   :52.05860
> |
```

i) Bar plot



Bar plot shows the average emission of individual groups goes on increasing with income levels.

ii) Box plot



Box plot shows means for each group are different.

To check if these groups have significantly different Average CO₂ emission, a formal test was carried out. Here Kruskal – Wallis rank test was used.

Kruskal – Wallis rank test serves as an alternative for ANOVA testing when Normality assumption fails. The data has observations that grouped into K groups. The null hypothesis assumes medians of all K groups are equal, while alternative hypothesis assumes otherwise. The test statistic H follows Chi-squared distribution with K-1 degrees of freedom and test rejects null hypothesis when H is large.

H_0 : All four groups have equal median of “Average CO₂ emission”

H_1 : Not all medians are equal

Testing was carried out on Minitab. The output of the test is as follows:

Kruskal-Wallis Test: Avg versus Income Group

Kruskal-Wallis Test on Avg

Income Group	N	Median	Ave Rank	Z
High Income	50	8.7553	125.2	9.04
Low Income	32	0.1448	18.4	-8.43
Lower Middle Income	35	0.8931	52.2	-3.86
Upper Middle Income	38	3.7917	89.8	1.87
Overall	155		78.0	

H = 125.96 DF = 3 P = 0.000

As p-value is 0 (Very close to 0), null hypothesis is rejected that is not all four medians of groups are equal.

This shows Average CO₂ emissions for years 2000-2014 differ significantly with income levels. Exploratory analysis shows countries in with High Income level and Low Income level have highest and lowest average CO₂ emissions respectively, among other groups.

Therefore, the group of High Income level countries like United States, Canada & Germany contributes highly to the average emission.

Time Series Analysis

1) Monthly Average Temperature Data

Exponential smoothing and ARIMA models are the two most widely used approaches to time series forecasting, and provide complementary approaches to the problem. While exponential smoothing models are based on a description of the trend and seasonality in the data, ARIMA models aim to describe the autocorrelations in the data.

i) Holt-Winters Method

The Holt-Winters seasonal method comprises the forecast equation and three smoothing equations — one for the level l_t , one for the trend b_t , and one for the seasonal component s_t , with corresponding smoothing parameters α , β and γ .

There are two variations to this method that differ in the nature of the seasonal component. The additive method is preferred when the seasonal variations are roughly constant through the series, while the multiplicative method is preferred when the seasonal variations are changing proportional to the level of the series. With the additive method, the seasonal component is expressed in absolute terms in the scale of the observed series, and in the level equation the series is seasonally adjusted by subtracting the seasonal component. Within each year, the seasonal component will add up to approximately zero. Here, we have considered additive method only.

We have considered data from 1951 to 2013. Holt-Winters additive method is applied by using ‘Forecast’ package in R.

Summary of the output is given on next page. 96 values i.e. 8 years forecast has been made.

```

Forecast method: Holt-Winters' additive method

Model Information:
Holt-Winters' additive method

Call:
hwCy = Temp, h = 96, seasonal = "additive")

Smoothing parameters:
  alpha = 0.0739
  beta  = 1e-04
  gamma = 1e-04

Initial states:
  l = 9.1625
  b = -0.0021
  s = 3.2609 5.6856 6.1446 5.2944 3.0874 0.1375
      -3.6145 -5.2245 -6.3313 -5.5429 -2.8291 -0.0681

sigma: 1.3787

      AIC      AICc      BIC
5242.015 5242.883 5319.933

Error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set -0.009767443 1.363378 1.022165 -1.320771 42.64614 0.7411483 0.2449752

```

One drawback of this method is that this method doesn't handle missing data properly. Hence, we have to consider the subset of our data which does not have any of the missing values.

In Table 1, we can see the performance of this method with respect to Forecasting. Point Forecasts are way off the mark, but 95% C.I. are fairly accurate. To see whether the performance improves by other model, we move to Seasonal ARIMA models in the next section.

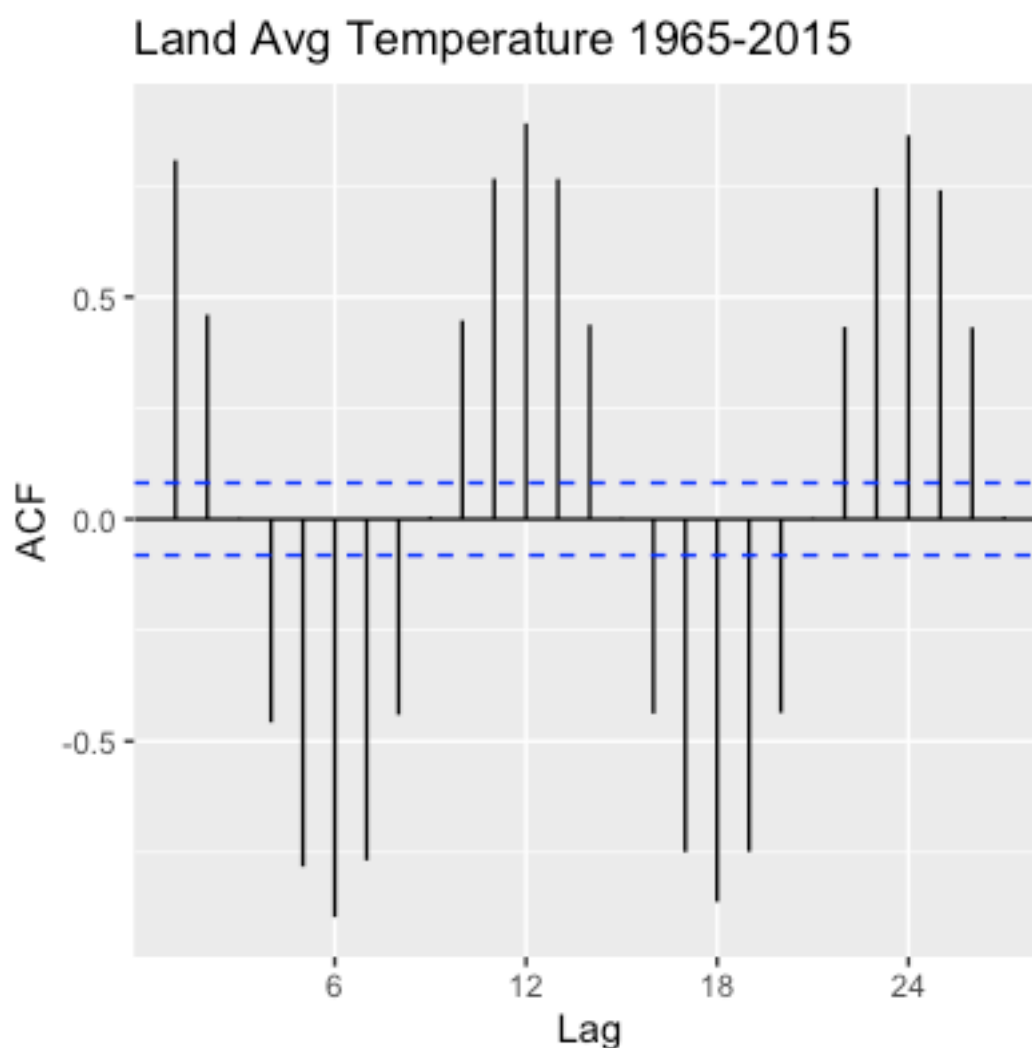
	Point Forecast	Low 95	High 95	Actual
Jan 2014	0.7254420	-1.9767982	3.427682	3.732
Feb 2014	1.8294965	-0.8801277	4.539121	3.500
Mar 2014	3.4370729	0.7200647	6.154081	6.378
Apr 2014	7.1859303	4.4615382	9.910322	9.589
May 2014	10.1332189	7.4014428	12.864995	12.582
Jun 2014	12.3374371	9.5982772	15.076597	14.335
Jul 2014	13.1852321	10.4386882	15.931776	14.873
Aug 2014	12.7235354	9.9696075	15.477463	14.875
Sep 2014	10.2964650	7.5351530	13.057777	13.091
Oct 2014	6.9644595	4.1957633	9.733156	10.330
Nov 2014	4.2005326	1.4244522	6.976613	6.713
Dec 2014	1.4835482	-1.2999168	4.267013	4.850
Jan 2015	0.6919535	-2.0989158	3.482823	3.881
Feb 2015	1.7960080	-1.0022460	4.594262	4.664
Mar 2015	3.4035844	0.5979453	6.209224	6.740
Apr 2015	7.1524418	4.3394174	9.965466	9.313
May 2015	10.0997304	7.2793204	12.920140	12.312
Jun 2015	12.3039487	9.4761528	15.131745	14.505
Jul 2015	13.1517436	10.3165616	15.986926	15.051
Aug 2015	12.6900469	9.8474783	15.532616	14.755
Sep 2015	10.2629765	7.4130209	13.112932	12.999
Oct 2015	6.9309710	4.0736282	9.788314	10.801
Nov 2015	4.1670441	1.3023136	7.031775	7.433
Dec 2015	1.4500597	-1.4220591	4.322178	5.518

ii) Seasonal ARIMA Models

A seasonal ARIMA model is formed by including additional seasonal terms in the ARIMA models. The seasonal part of the model consists of terms that are similar to the non-seasonal components of the model, but involve backshifts of the seasonal period. For example, an ARIMA(1,1,1)(1,1,1)₄ model (without a constant) is for quarterly data, and can be written as

$$(1 - \phi_1 B) (1 - \Phi_1 B^4)(1 - B)(1 - B^4)y_t = (1 + \theta_1 B) (1 + \Theta_1 B^4)\varepsilon_t.$$

ACF of the Time Series of Land Temperatures is shown below.



Seasonal ARIMA model is fitted using 'Forecast' package in R.

The `auto.arima()` function in R uses a variation of the Hyndman-Khandakar algorithm (Hyndman and Khandakar, 2008), which combines unit root tests, minimisation of the AICc and MLE to obtain an ARIMA model.

SARIMA (1,0,0)(2,1,0)12 is found to be most suitable model for our data.

Series: Data_TS

ARIMA(1,0,0)(2,1,0)[12]

Coefficients:

	ar1	sar1	sar2
	0.2958	-0.6042	-0.2601
s.e.	0.0394	0.0396	0.0396

sigma^2 estimated as 2.536: log likelihood=-1116.13

AIC=2240.27 AICc=2240.33 BIC=2257.85

Residual diagnostics are carried out. From Ljung-Box test, we can see that model is acceptable at 1% l.o.s.

Ljung-Box test

```
data: Residuals from ARIMA(1,0,0)(2,1,0)[12]
Q* = 34.02, df = 21, p-value = 0.03606
```

```
Model df: 3. Total lags used: 24
```

From table below, for first few forecasts SARIMA model is performing better than Holt-Winters. But, as we move away from the present, FMSE starts increasing for SARIMA.

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95	Actual
Jan 2014	3.212130	1.13911345	5.285147	0.04172477	6.382535	3.732
Feb 2014	3.666320	1.50424240	5.828398	0.35970761	6.972933	3.500
Mar 2014	5.060636	2.89091556	7.230356	1.74233514	8.378936	6.378
Apr 2014	7.483360	5.31297055	9.653750	4.16403572	10.802685	9.589
May 2014	11.169146	8.99869726	13.339594	7.84973133	14.488560	12.582
Jun 2014	13.706655	11.53620180	15.877109	10.38723314	17.026077	14.335
Jul 2014	14.492738	12.32228382	16.663192	11.17331491	17.812161	14.873
Aug 2014	14.496803	12.32634863	16.667257	11.17737971	17.816225	14.875
Sep 2014	12.016150	9.84569642	14.186604	8.69672749	15.335573	13.091
Oct 2014	8.911102	6.74064754	11.081555	5.59167862	12.230524	10.330
Nov 2014	5.509120	3.33866627	7.679574	2.18969734	8.828543	6.713
Dec 2014	2.264490	0.09403591	4.434944	-1.05493302	5.583913	4.850
Jan 2015	3.330902	1.01184925	5.649956	-0.21578338	6.877588	3.881
Feb 2015	3.840405	1.50876114	6.172049	0.27446351	7.406346	4.664
Mar 2015	5.481069	3.14832354	7.813815	1.91344264	9.048696	6.740
Apr 2015	7.465109	5.13226640	9.797951	3.89733432	11.032883	9.313
May 2015	11.184735	8.85188436	13.517586	7.61694778	14.752522	12.312
Jun 2015	13.583447	11.25059554	15.916298	10.01565857	17.151235	14.505
Jul 2015	14.553250	12.22039843	16.886101	10.98546143	18.121039	15.051
Aug 2015	14.468925	12.13607311	16.801776	10.90113610	18.036713	14.755
Sep 2015	11.994431	9.66157987	14.327283	8.42664287	15.562220	12.999
Oct 2015	8.893838	6.56098627	11.226689	5.32604926	12.461626	10.801
Nov 2015	5.502677	3.16982503	7.835528	1.93488802	9.070465	7.433
Dec 2015	2.441728	0.10887615	4.774579	-1.12606086	6.009516	5.518

As both of these methods are not useful for long term forecast, we will do analysis on another time series i.e. Annual Global Temperature. As this is an annual time series, there is no question of seasonality.

2) Annual Global Temperature Data

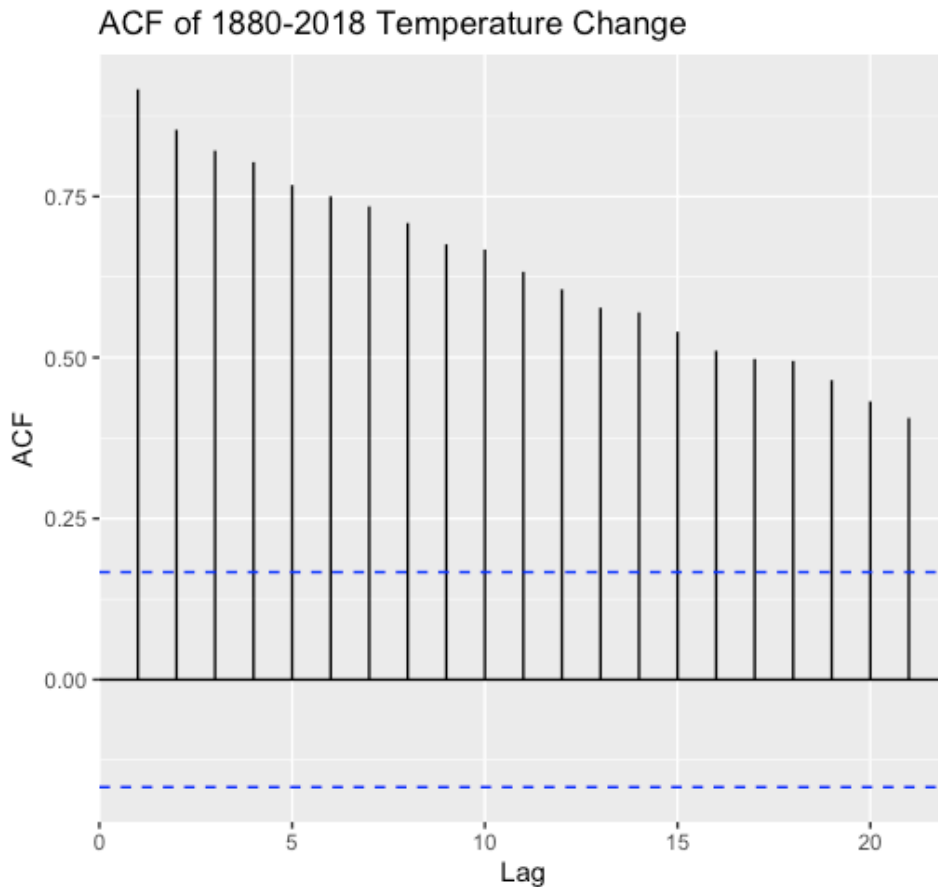
ARIMA Modelling :

If we combine differencing with autoregression and a moving average model, we obtain a non-seasonal ARIMA model. ARIMA is an acronym for AutoRegressive Integrated Moving Average (in this context, “integration” is the reverse of differencing). The full model can be written as

$$\underbrace{(1 - \phi_1 B - \dots - \phi_p B^p)}_{\uparrow \text{AR}(p)} \underbrace{(1 - B)^d}_{\uparrow d \text{ differences}} y_t = c + \underbrace{(1 + \theta_1 B + \dots + \theta_q B^q)}_{\uparrow \text{MA}(q)} \varepsilon_t$$

We considered Annual Global Average Temperature Data, freely available on NASA Data Portal.

ACF of the 1961-2018 Data is given below.



Using *auto.arima()*, ARIMA(1,1,3) model is obtained as most suitable.

Series: Data_TS
ARIMA(1,1,3) with drift

Coefficients:

	ar1	ma1	ma2	ma3	drift
	-0.9446	0.6299	-0.5758	-0.338	0.0076
s.e.	0.0511	0.0938	0.0843	0.080	0.0032

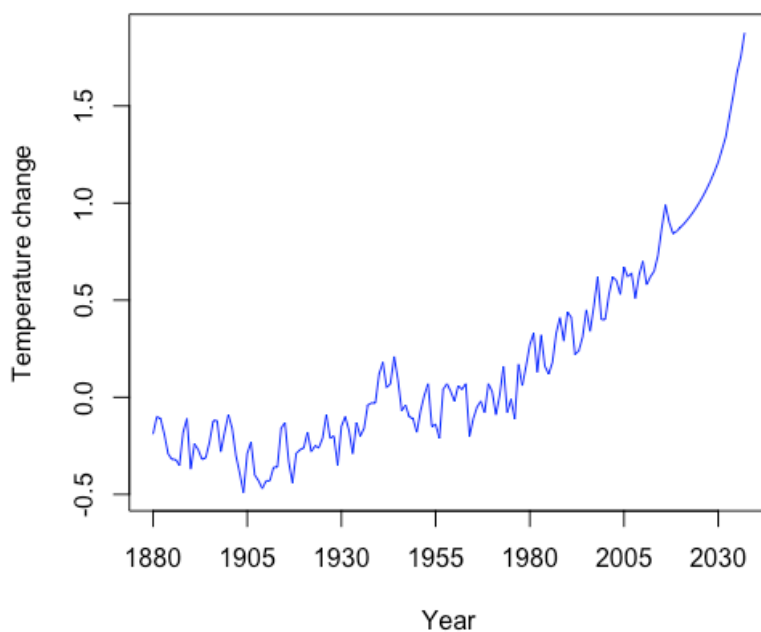
sigma^2 estimated as 0.01038: log likelihood=120.72
AIC=-229.45 AICc=-228.8 BIC=-211.93

Ljung-Box test

data: Residuals from ARIMA(1,1,3) with drift
Q* = 3.9577, df = 5, p-value = 0.5555

Model df: 5. Total lags used: 10

From Ljung- Box test, Model is clearly acceptable as p-value is very high.



Forecasts till 2038 are made using iterative forecasting algorithm. From graph, we can clearly see that the increasing trend is expected to continue in the future.

Year	Forecast
2018	0.8428
2020	0.8753
2022	0.9187
2024	0.9722
2026	1.0372
2028	1.1158
2030	1.2109
2032	1.3409
2034	1.5552
2036	1.7527
2038	1.8777

We are forecasting about 1°C increase in Earth's temperature in next 20 years. By this rate, we would cross the threshold of 2°C (set by Paris agreement) before 2040.

Cross Correlation & Causality between temperature & CO₂emissions

Granger test for causality:

Granger test for causality is used to determine if we can forecast one time series with the help of other time series' available to us. If we have two time series, say, **X&Y**, & if **X** Granger-cause **Y** then past values of **X** are helpful in forecasting future values of **Y**. In our case, we had yearly time series data on temperature & CO₂ emissions. We wanted to check how many past years' CO₂ emission is useful in forecasting temperature. The null hypothesis was lag k of CO₂ emissions do not Granger-cause temperature. We have two models as:

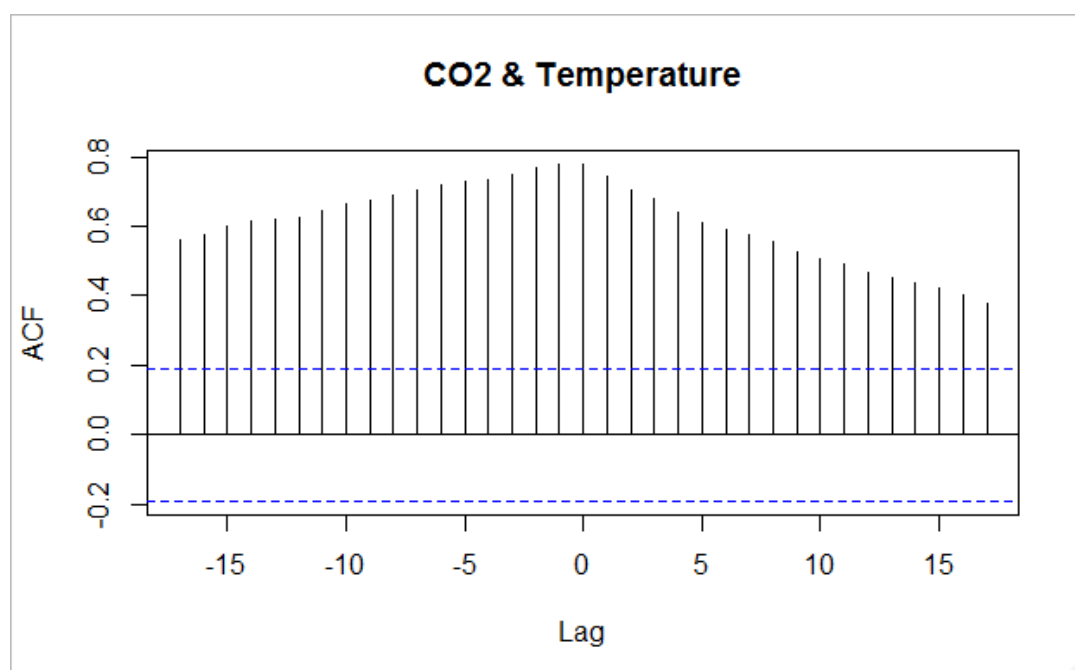
Model 1: $y(t) = \sum_{i \geq 1} a_i y(t-i) + \sum_{i=1}^k b_i x(t-i) + c_1 + v_1(t)$

Model 2: $y(t) = \sum_{i \geq 1} a_i y(t-i) + c_2 + v_2(t)$

We calculate F-statistic as: $F = \frac{(ESS_{Model2} - ESS_{Model1})/q}{ESS_{Model1}/(n-k)}$

where q is the difference in the parameters of two models. We reject the null hypothesis at $\alpha\%$ level of significance if the test statistic value is greater than $F_{(1-\alpha),q,(n-k)}$.

We first calculated cross correlation between CO₂& temperature using Data 1 & Data 9 which is as shown below.



We see that there is positive correlation between CO₂ emissions & temperature. Correlations from lag -17 to 17 are significant & hence there is lagged effect of CO₂ on temperature.

Granger's Test for causality:

We tested whether CO₂ emissions are useful in forecasting temperature. We used Granger's test of causality to test this hypothesis. We also tested if CO₂ emissions & temperature are instantaneously related.

Causality between Temperature & CO₂:

Lag	F statistic	Table value	p-value	Decision
1	11.409	1.592445	0.001039	Reject H ₀ .
2	4.7632	1.602355	0.01061	Reject H ₀ .
3	2.2837	1.61277	0.08394	Reject H ₀ .
4	1.4181	1.623736	0.2342	Fail to reject H ₀ .

(At 1% level of significance)

We reject the null hypothesis for lags 1, 2 & 3. There is Granger cause relationship between CO₂ emissions & temperature. Hence, we may be able to forecast temperature based on CO₂ emissions.

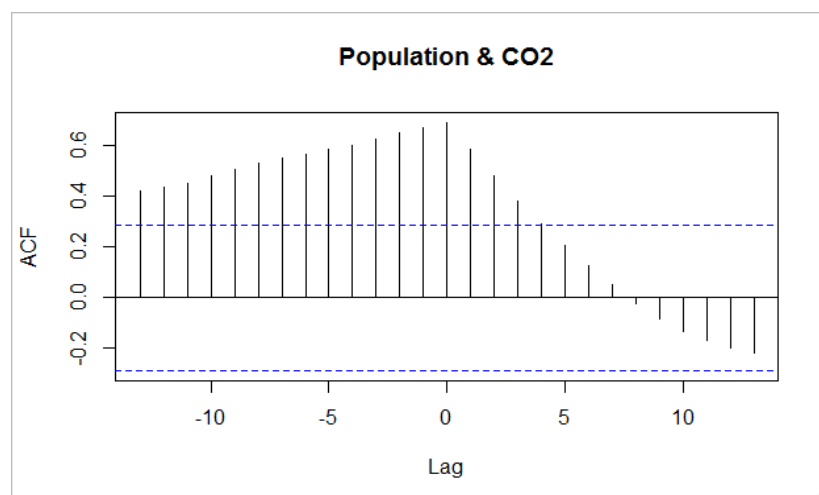
Test for instantaneous causality:

We tested if there is instantaneous causal relationship between CO₂ emissions & temperature. The observed value of chi-square test statistic was found to be 4.1498. This is greater than chi-square cut off at 5% level of significance which is 3.841459 but less than chi-square cut off at 1% which is 6.634897. p-value was 0.04164. Thus, we may say that there is instantaneous causality between CO₂ emissions & temperature.

Relationship between CO₂ emissions & other variables

1. Population

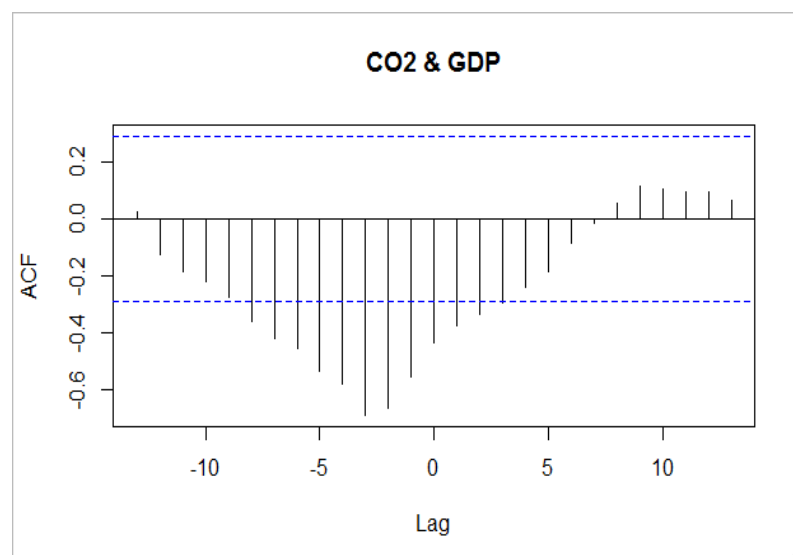
We used Data 1 & Data 4 for computing cross correlations.



We see that there is positive correlation between CO₂ emissions & population. Correlations from lag -13 to 3 are significant & hence there is lagged effect of population on CO₂ emissions.

2. GDP growth

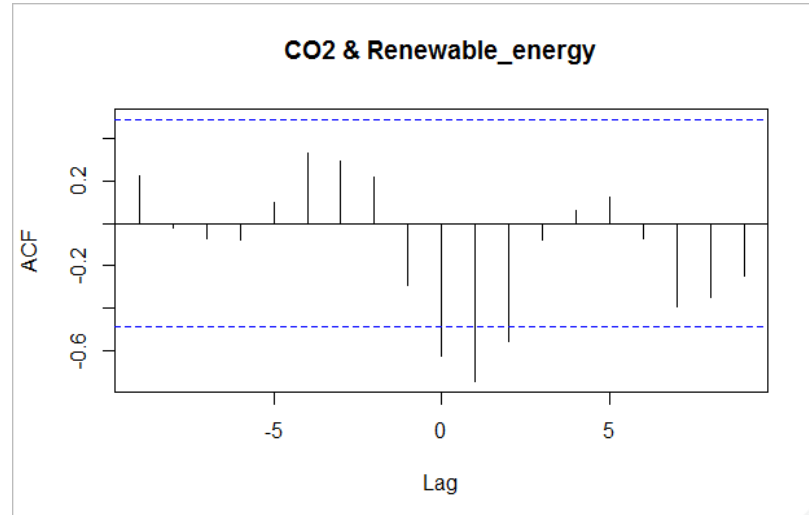
We used Data 1 & Data 5 for computing cross correlations here.



There is negative correlation between CO₂ emissions & population. Correlations from lag -8 to 2 are significant & hence there is lagged effect of GDP growth on CO₂ emissions.

3. Renewable energy consumption (% of final energy consumption)

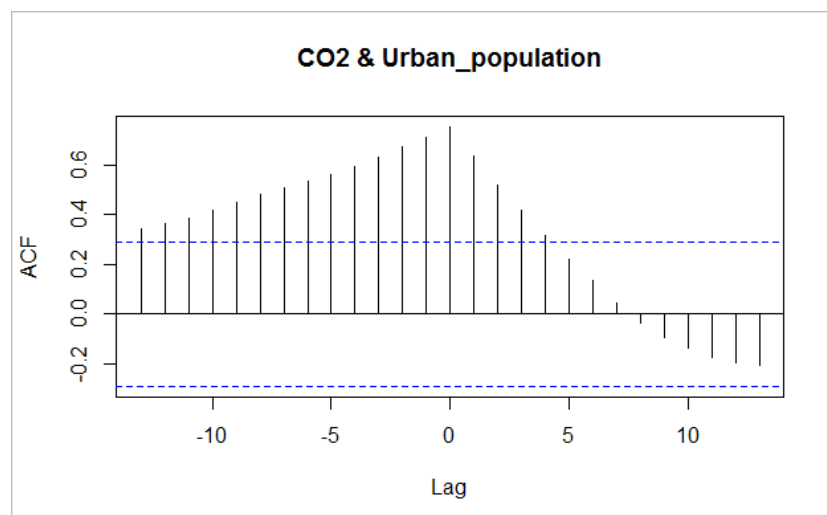
We used Data 1 & Data 6 to compute cross correlations.



We see that there is negative correlation between CO₂ emissions & consumption of renewable energy. Correlations of lag 0, 1 & 2 are significant & hence there may be lagged effect of consumption of renewable energy on CO₂ emissions.

4. Urban population (% of total population)

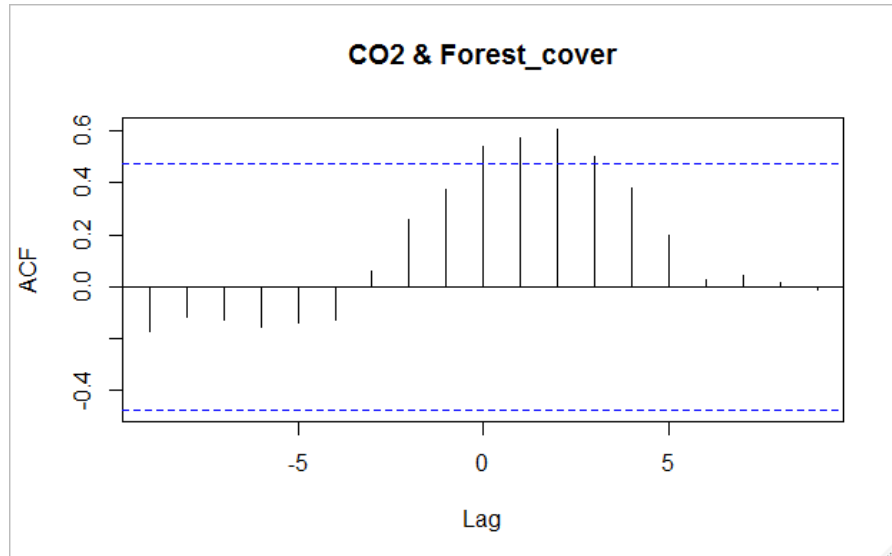
We used Data 1 & Data 7 to compute cross correlations.



We see that there is positive correlation between CO₂ emissions & urban population. Correlations from lag -13 to 4 are significant & hence there is lagged effect of urban population on CO₂ emissions.

5. Forest cover (in hectares)

We used Data 1 & Data 8 to compute cross correlations.



We see that there is positive correlation between CO₂ emissions & forest cover. Correlations from lag 0 to 3 are significant & hence there is lagged effect of forest cover on CO₂ emissions.

Test for instantaneous causality:

Variable	Chi-square Test statistic	p-value	Decision
Population	0.28206	0.5954	Fail to reject H ₀ .
GDP growth	8.9248	0.002813	Reject H ₀ .
Urban population (% of total population)	0.24982	0.6172	Fail to reject H ₀ .
Forest cover (in hectares)	6.0000	0.01431	Reject H ₀ .

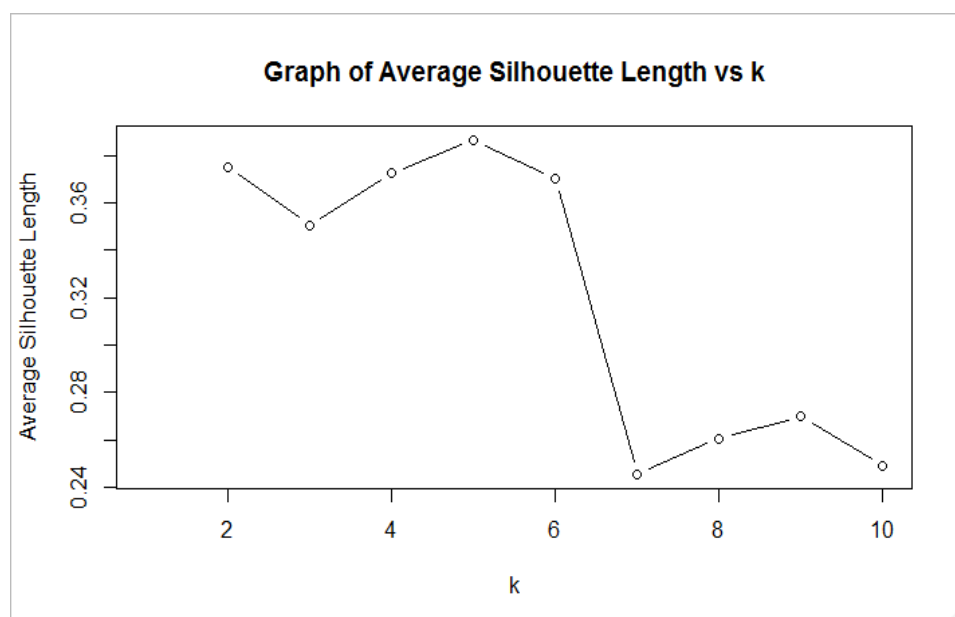
(Chi-square table value for 1 d.f. at 95% level of significance is 3.841459)

For variables population & forest cover, we do not reject H₀. Hence, there is instantaneous causality between population & CO₂ emissions and forest cover & CO₂ emissions.

Time Series Clustering using PAM

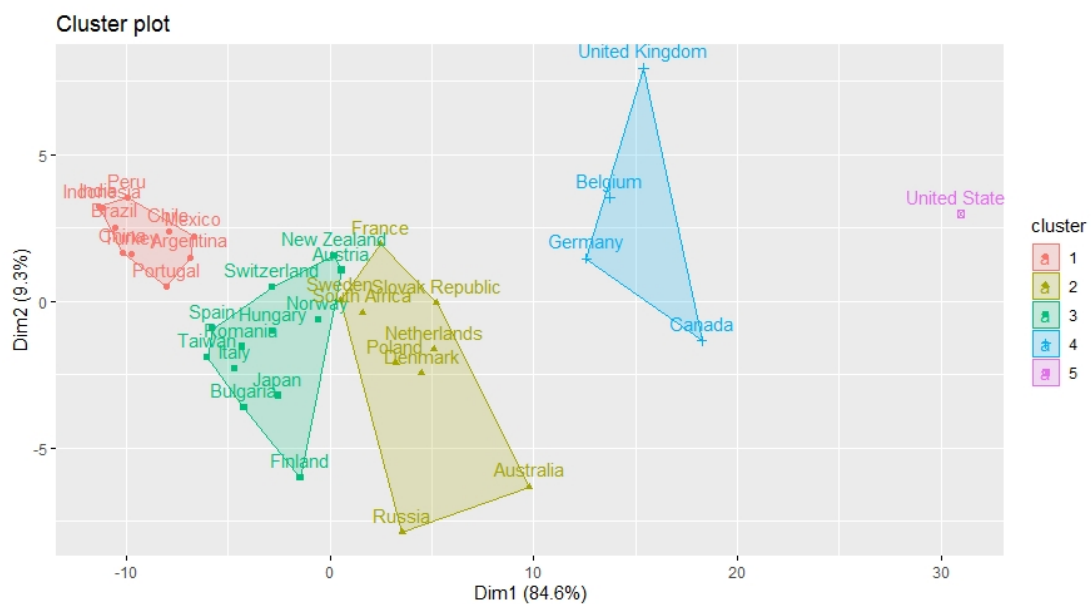
We were interested to see if we can cluster countries so that we have countries following similar patterns in CO₂ emissions in one cluster. Using this, we may be able to devise strategies for all the involved countries by studying just few. We used PAM to cluster countries with same patterns in CO₂ emissions & energy consumption over the years. PAM or Partitioning Around Medoids is reminiscent to k-means clustering. It also attempts to divide the data into different clusters. But it has several advantages over k-means clustering. It is more robust to outliers & we do not always need the actual data. In addition, it does not require a dissimilarity measure to be based on Euclidean distance or Manhattan distance. Due to these advantages, we chose to use PAM to find out countries having similar behaviour.

1. For time series clustering based on CO₂ emission, we considered Data 1 for analysis. We calculated dissimilarities between two countries using dissimilarity function based on Auto-Correlation Functions (ACF) of time series. Range of this function was from 0 to 2. Lower values or values close to 0 suggested that two series had similar ACF while values close to 2 suggested that two series had different ACF. Hence, we constructed a dissimilarity matrix for 36 countries. Optimal number of clusters, which was found to be 5, was found using Silhouette statistic as a measure of homogeneity. The 5 clusters are as follows:



We see that average silhouette length for $k = 5$ is maximum. Hence, 5 medoid clustering will give us most homogenous clusters.

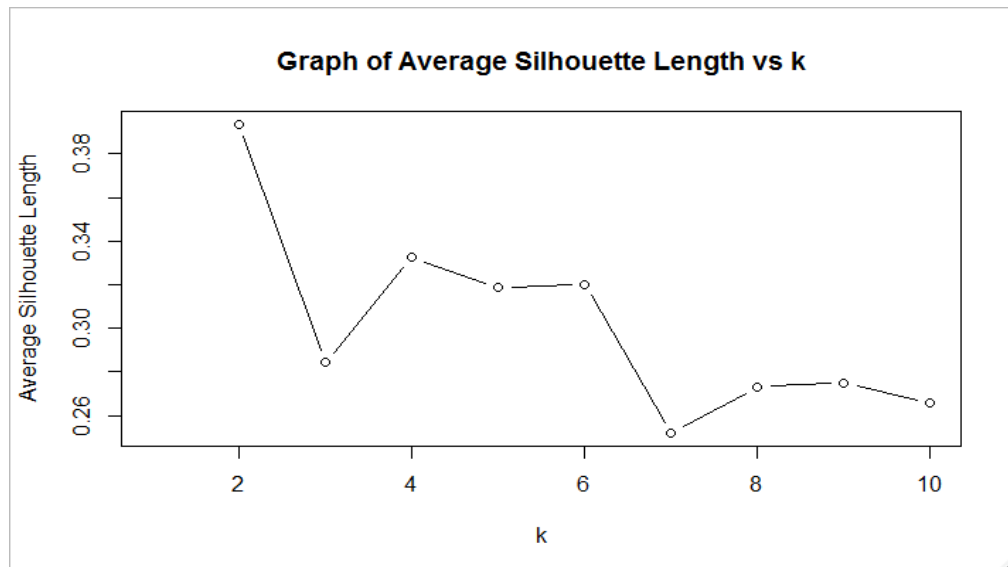
Clustering is done as shown:



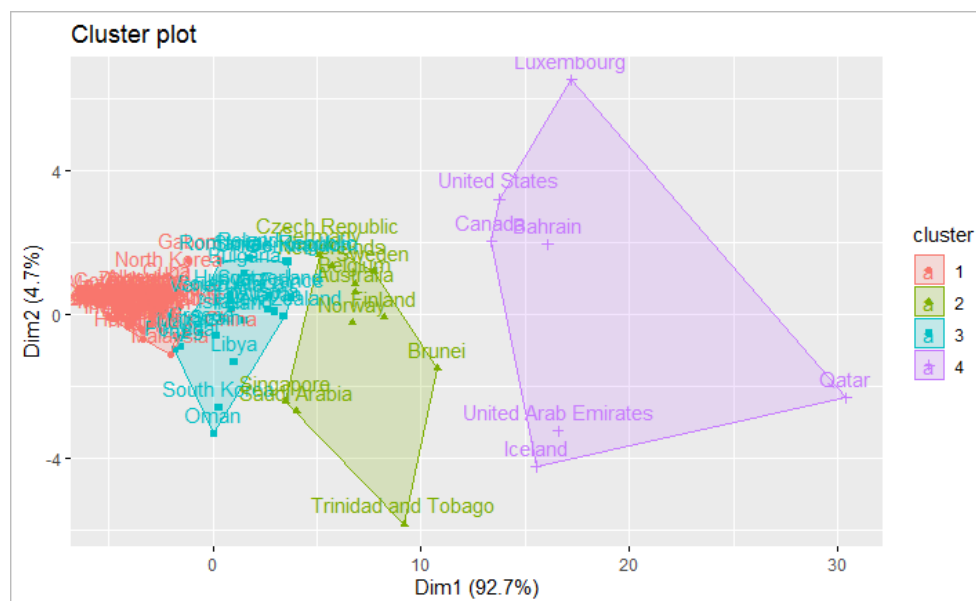
We see the 5 clusters which can be described as follows:

- Cluster 1 contains developing countries such as India, Indonesia, and Brazil etc. These are the countries that have large population & have scope to develop their economies further.
- Cluster 2 & 3 contains countries which are not the biggest economies in the world (except Japan & France) but are amongst the developed countries.
- Cluster 4 contains Canada, the United Kingdom, Germany & Belgium. These are some of the largest economies in the world.
- Cluster 5 contains only the United States which is far away from all the clusters & hence need to study separately.

2. For time series clustering based on energy consumption, we considered Data 2 for analysis. Using similar algorithm, we clustered countries showing similar trends in energy consumption. We experimented with different dissimilarity measures & found that ‘ACF’ dissimilarity measure gave us best clusters. Although, optimal number of clusters was 2, we chose 4 clusters to increase interpretability of clusters. Average silhouette length for 2 clusters close to 0.38 while for 4 clusters it is 0.34. Hence, we do not lose much of homogeneity for the sake of interpretability.



Clusters formed using 4 medoid clustering are as follows:



Clusters can be interpreted as follows:

- Cluster 1 includes most of the developing nations such as India, Ivory Coast & Brazil. Many of the countries in this cluster are low income group countries from Africa, Asia & South America.
- Cluster 2 includes countries such as Germany, Australia & Norway. These are the developed countries who are promoting use of renewable energy.

- Cluster 3 includes developed European nations along with some other countries like South Africa, South Korea & Iran. These countries have not increased their per capita energy consumption much over the years.
- Cluster 4 contains high income group developed countries whose per capita energy consumption is very high.

Nation-wise CO₂ Emissions: Time Series Analysis

While it's still technically possible to limit global warming to within the limits set by the Paris Agreement, rising carbon emissions present a grave threat to the planet. Less than 200 years ago there was hardly any CO₂ in the atmosphere. In the mid-1700s, the Industrial Revolution, driven by the burning of fossil fuels, started a process that would have serious repercussions for the world we live in today.

For this analysis, we have considered per capita CO₂ Emissions of 8 countries from 1881-2017. ARIMA modelling have been done on the time series of each country¹. The results are obtained as follows :

1) Developing Nations

In this set, we have considered four developing economies China, India, Brazil and Egypt. Notably, Egypt and India are ranked very high in the Climate Performance Index by IPCC.

Country	Model	Current Value	Forecast of 2030	%Change
China	ARIMA(2,1,2) with drift	6.98	16.08	130.37
India	ARIMA(0,2,1)	1.84	3.02	64.13
Egypt	ARIMA(1,1,0) with drift	2.24	4.73	111.16
Brazil	ARIMA(0,1,0) with drift	2.27	3.14	38.33

From this table, we see that all these countries have increasing trend in per capita CO₂ emissions². For China and Egypt, we are forecasting more than twice increase in CO₂ emissions. For India, expected increase is around 64%.

¹Forecast package from R is used

²Visualization can be seen on Pg. 14

2) Developed Nations

In this set, four developed economies – US, UK, Japan and Sweden are considered. All of these countries are economically stable and have high Human Development Index (HDI).

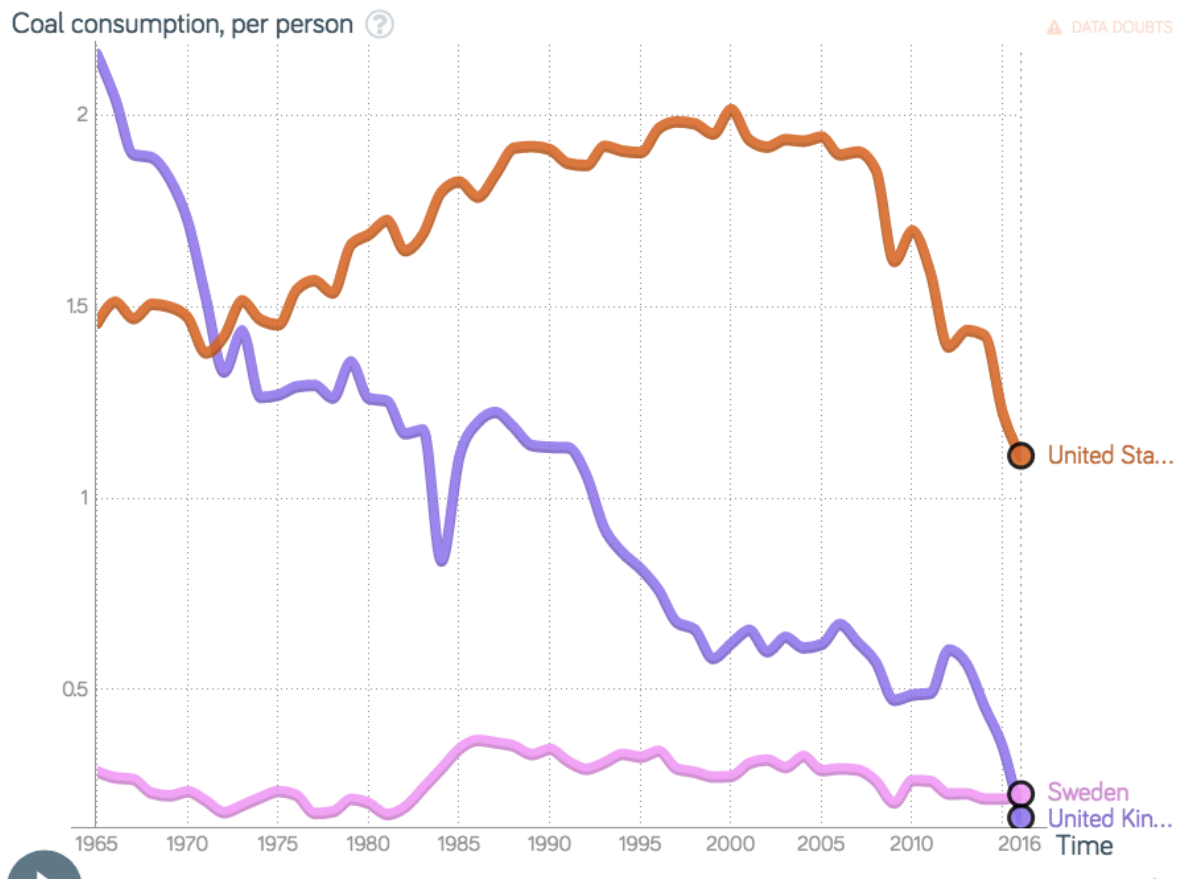
Country	Model	Current Value	Forecast of 2030	% Change
UK	ARIMA(5,2,1)	5.81	4.03	-30.64
USA	ARIMA(0,2,2)	16.24	14.44	-11.08
Sweden	ARIMA(2,2,3)	4.18	3.40	-18.66
Japan	ARIMA(0,2,2)	9.45	10.18	7.72

US, UK and Sweden are set to reduce their per capita emissions significantly till 2030. Notably, according to our forecast, UK will reduce its emissions by 30% in next 10 years. We need to study climate policies of these countries, which are causing steady decrease in their CO₂ emissions.³

Why UK's CO₂ Emissions have fallen?

- UK's CO₂ emissions are about 38% lower than 1990, and projected to reduce further in next 10 years.
- The largest driver has been a cleaner electricity based on gas and renewables instead of coal.
- Another factor is reduced fuel consumption by business and country.
- Overall emissions in 2017 were 50% lower that they would have been without these changes.

³See Pg. 15



UK is showing decreasing trend in per capita coal consumption from 1965. For US, trend was somewhat stable till 2005, but it started decreasing at the turn of this decade.

Limitations and Scope

- As there is no single portal devoted to Climate Change Data, we have to rely on different sources and agencies. Hence, the reliability of this work is dependant upon the credibility of the sources.
- Majority of our work is restricted only to Univariate Time Series.
- We have to restrict our study on only those years for which data on all the variables are available.
- More advanced models like Fuzzy Expert System, General Linear Models, Digital Image Resolution can be implemented in the future.
- We can simulate 3-4 possible scenarios for the future and forecast rise/fall in global temperature in each scenario. This requires high computing power and thorough understanding of Machine Learning.

Strategies to be Implemented

We are proposing five point agenda for fighting with the Global Warming:

1. The inequality in global emissions lies at the heart of why international agreement on climate change has so contentious. Developed nations have to focus on reducing CO₂ emissions while looking for sustainable development of their economies. Climate policies of UK and Sweden need to be studied and should be implemented in all developed nations.
2. For developing nations like India, more and more funds should be invested in Renewable Energy. At least 40% of total energy consumption should be from Renewable energy.
3. Instead of craving to have 7-8% GDP growth and then in consequence have 6-7% increase in CO₂ emissions, we should target to have 3-4% GDP growth with 1-2% (Ideally zero) increase in CO₂ emissions. This will lead to sustainable growth of our economy.
4. Tremendous increase in urban population is a cause of concern. We should try to diversify job and business opportunities in all parts of the country.
5. All the countries should at least try to maintain their forest cover. This can be done by subscribing to and implementing Paris agreement.

Concluding Remarks

Global Warming is the reality and has always been a sensitive subject to handle. The Earth's temperature is projected to increase by 2°C in next 20-25 years, which is a concerning fact for all of the humankind. Ignorance at this stage will have drastic implications in the future.

Some of the highly developed nations like UK, USA, Sweden have similar trend of CO₂ emissions – Increasing until the last decade and then a decrease. In particular, huge reduction in CO₂ emissions has been a great achievement for UK. But overall climatic performance of highincome level countries is not satisfactory. Some major developing nations like India, Brazil are also set to increase their CO₂ emissions drastically in next two decades.

Developed countries should invest in other countries so as to achieve global efficiency. Investment in research and implementation of Renewable Energy must be encouraged. In the last couple of years Government has been taking strides in advancement of Green Energy. Urban population and controlling the rate of forest cover loss are some of the factors that need special attention and efforts.

Simple statistical models and tools gave good results on the data. We would like to extend this project in the future as we learn some more advanced statistical techniques.

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